Dolby® Multichannel Amplifier

User Manual

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PRODUCT MODEL:

THIS DOCUMENTATION APPLIES TO MODEL CID1001

Regulatory Notices

United States (FCC)

This equipment complies with Part 15 of the FCC rules. Operation is subject to the following conditions:

1. This device may not cause harmful interference.
2. This device must accept any interference received, including interference that may cause undesired operation.

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception (which can be determined by turning the equipment off and on), the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment to an outlet on a circuit that is different from the outlet to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

Any changes or modifications to this product could void the authorization provided to the user to operate this device.

Properly shielded and grounded cables and connectors must be used for connection to other accessories and/or peripherals in order to meet FCC emission limits.

Canada

This Class B digital apparatus complies with Canadian ICES-003.

Cet appareil numérique de la classe B est conforme à la norme NMB-003 du Canada.

European Union

The power supply on this product has power factor correction. However, we recommend that you notify your supply company or authority prior to connecting.

A Declaration of Conformity is available upon request.

Important Safety Instructions

1. Read and follow these instructions.
2. Keep these instructions.
3. Heed all WARNINGS.
4. WARNING: Do not use this apparatus near water.
5. WARNING: To reduce the risk of fire or electric shock, do not expose this apparatus to rain or moisture.
6. WARNING: There are high voltages present within the product enclosure. To prevent electric shock, do not remove the top cover or attempt to service the unit in any way. There are no user-serviceable parts inside the unit. Refer all servicing and troubleshooting to qualified service personnel only.
7. Clean only with dry cloth.
8. Do not install near any heat sources such as radiators, heat registers, stoves, or other apparatus (including amplifiers) that produce heat.
9. No naked flame sources, such as lighted candles, should be placed on the apparatus.
10. Protect the power cord from being walked on or pinched, particularly at plugs, convenience receptacles, and the point where it exits from the apparatus.
11. Only use attachments/accessories specified by the manufacturer.
12. Unplug this apparatus when unused for long periods of time.
13. Refer all servicing to qualified service personnel. Servicing is required when the apparatus has been damaged in any way, such as the power-supply cord or plug is damaged, liquid has been spilled or objects have fallen into the apparatus, the apparatus has been exposed to rain or moisture, does not operate normally, or has been dropped.
14. Do not expose the apparatus to dripping or splashing, and no objects filled with liquids, such as vases, should be placed on the apparatus.
15. WARNING: There are high voltages present within the product enclosure. Troubleshooting must be performed by a trained technician. To reduce the risk of electric shock, do not attempt to service this equipment unless you are qualified.
16. Do not defeat the safety purpose of the polarized or grounding-type plug. A polarized plug has two blades with one wider than the other. A grounding-type plug has two blades and a third grounding prong. The wide blade or the third prong is provided for your safety. If the provided plug does not fit into your outlet, consult an electrician for replacement of the obsolete outlet.
17. This apparatus must be earthed (grounded) by connecting to a correctly wired and earthed power outlet.
18. Ensure that your mains supply is in the correct range for the input power requirement of the unit.
19. This equipment is designed to mount in a suitably ventilated 19” rack; ensure that any ventilation slots in the unit are not blocked or covered.
20. To avoid exposure to dangerous voltages and to avoid damage to the unit, do not connect the rear-panel Ethernet port to telephone circuits.
21. As the colors of the cores in the mains lead may not correspond with the colored markings identifying the terminals in your plug, proceed as follows:
    - The green or green and yellow core must be connected to the terminal in the plug identified by the letter E, or by the earth symbol ⏚, or colored green, or green and yellow.
    - The blue and brown cores must be connected to the power pins in the mains plug according to local code.

Product End-Of-Life Information

This product was designed and built by Dolby Laboratories to provide many years of service and is backed by our commitment to provide high-quality support. When it eventually reaches the end of its serviceable life, it should be disposed of in accordance with local or national legislation. For current information please visit our website: https://www.dolby.com/about/legal/weee-packaging-battery/.

Warning and Safety Symbols

This symbol that appears in this manual is intended to alert the user to the presence of uninsulated “dangerous” voltage within the product’s enclosure that may be of sufficient magnitude to constitute a risk of electric shock to persons.

This symbol that appears in this manual is intended to alert the user to the presence of important safety operating and maintenance instructions.

This symbol that appears on the unit rear panel and in this manual is intended to alert the user to the presence of uninsulated “dangerous” voltage within the product’s enclosure that may be of sufficient magnitude to constitute a risk of electric shock to persons.

This symbol that appears on the unit rear panel is intended to alert the user to the presence of important safety operating and maintenance instructions.

This symbol that appears on the unit rear panel is intended to alert the user that high voltage is present at the speaker output terminals and that these terminals should be handled and connected only by an authorized (trained) technician. Use Class-2 wiring for all speaker connections.

IMPORTANT SAFETY NOTICE

GB

To ensure safe operation and to guard against potential shock hazard or risk of fire, the following **must** be observed:

- Ensure that your mains supply is in the correct range for the input power requirement of the unit.
- Ensure **fuses** fitted are the **correct rating and type** as marked on the unit.
- The unit **must be earthed** by connecting to a correctly wired and **earthed** power outlet.
- The **power cord** supplied with this unit must be wired as follows:

Live—Brown   Neutral—Blue   Earth—Green/Yellow

IMPORTANT – NOTE DE SECURITE

F

Pour vous assurer d'un fonctionnement sans danger et de prévenir tout choc électrique ou tout risque d'incendie, veillez à ob les recommandations suivantes.

- Le selecteur de tension doit être placé sur la valeur correspondante à votre alimentation réseau.
- Les fusibles doivent correspondre à la valeur indiquée sur le materiel.
- Le materiel doit être correctement relié à la terre.
- Le cordon secteur livré avec le materiel doit être cablé de la manière suivante:

Phase—Brun   Neutre—Bleu   Terre—Vert/Jaune

WICHTIGER SICHERHEITSHINWEIS

D

Für das sichere Funktionieren der Geräte und zur Unfallverhütung (elektrischer Schlag, Feuer) sind die folgenden Regeln unbedingt einzuhalten:

- Der Spannungswähler muß auf Ihre Netzspannung eingestellt sein.
- Die Sicherungen müssen in Typ und Stromwert mit den Angaben auf dem Gerät übereinstimmen.
- Die Erdung des Gerätes muß über eine geerdete Steckdose gewährleistet sein.
- Das mitgelieferte Netzkabel muß wie folgt verdrahtet werden:

Phase—braun   Nulleiter—blau   Erde—grün/gelb

NORME DI SICUREZZA – IMPORTANTE

I

Per una perfetta sicurezza ed al fine di evitare eventuali rischi di scossa elettrica o d'incendio vanno osservate le seguenti misure di sicurezza:

- Assicurarsi che il selettore di cambio tensione sia posizionato sul valore corretto.
- Assicurarsi che la portata ed il tipo di fusibili siano quelli prescritti dalla casa costruttrice.
- L'apparecchiatura deve avere un collegamento di messa a terra ben eseguito; anche la connessione rete deve avere un collegamento a terra.
- Il cavo di alimentazione a corredo dell'apparecchiatura deve essere collegato come segue:

Filo tensione—Marrone   Neutro—Blu   Massa—Verde/Giallo

AVISO IMPORTANTE DE SEGURIDAD

E

Para asegurarse un funcionamiento seguro y prevenir cualquier posible peligro de descarga o riesgo de incendio, se han de observar las siguientes precauciones:

- Asegúrese que el selector de tensión esté ajustado a la tensión correcta para su alimentación.
- Asegúrese que los fusibles colocados son del tipo y valor correctos, tal como se marca en la unidad.
- La unidad debe ser puesta a tierra, conectándola a un conector de red correctamente cableado y puesto a tierra.
- El cable de red suministrado con esta unidad, debe ser cableado como sigue:

Vivo—Marrón   Neutro—Azul   Tierra—Verde/Amarillo

VIKTIGA SÄKERHETSÅTGÄRDER!

S

För att garantera säkerheten och gardera mot eventuell elchock eller brandrisk, måste följande obas:

- Kontrollera att spänningsväljaren är inställd på korrekt nätspänning.
- Konrollera att säkringarna är av rätt typ och för rätt strömstyrka så som anvisningarna på enheten föreskriver.
- Enheten måste vara jordad genom anslutning till ett korrekt kopplat och jordat el-uttag.
- El-sladden som medföljer denna enhet måste kopplas enligt foljande:

Fas—Brun   Neutral—Blå   Jord—Grön/Gul

BELANGRIJK VEILIGHEIDS-VOORSCHRIFT:

NL

Voor een veilig gebruik en om het gevaar van electrische schokken en het risico van brand te vermijden, dienen de volgende regels in acht te worden genomen:

- Controleer of de spanningscarousseel op het juiste Voltage staat.
- Gebruik alleen zekeringen van de aangegeven typen en waarden.
- Aansluiting van de unit alleen aan een geaarde wandcontactdoos.
- De netkabel die met de unit wordt geleverd, moet als volgt worden aangesloten:

Fase—Bruin   Nul—Blauw   Aarde—Groen/Geel

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Introduction

The Dolby® Multichannel Amplifier is an advanced high-density amplifier in a compact 4U rackmount chassis. The Class D amplification topology is designed to deliver high-performance audio quality on every channel.

The Dolby Multichannel Amplifier is designed for reliability. It includes a custom-built power supply with intrinsic redundancy, power sharing, operational monitoring, and fault detection. The power supply design enables the system to operate from 100 to 240 VAC, up to a 20-amp service without tripping the AC mains circuit breaker. The Dolby Multichannel Amplifier automatically detects maximum and net power usage and certain operational and environmental conditions, and adjusts channel gains based on power supply conditions, load conditions, and fault conditions.

The Dolby Multichannel Amplifier receives digital audio through the Dolby Atmos® Connect bitstream interface, which is included on Dolby products such as the Dolby Cinema Processor CP950 (product model CID1019), Dolby Atmos Cinema Processor CP950A, the Dolby Atmos Cinema Processor CP850, and the Dolby Integrated Media Server IMS3000 with an internal audio processor enabled (product model CID1002).

The Dolby DMA16302 and DMA24302 include an analog input card, which enables these amplifiers to receive audio from a Dolby CP750 or a similar device. Earlier units can receive audio from a CP750 or other audio processor via a Yamaha Tio1608-D analog-to-digital converter. For information on this type of configuration, log into https://customer.dolby.com/cinema/ and download the *Dolby CP750 7.1 audio for the Dolby Multichannel Amplifier Installation Manual*.

You can also interconnect additional Dolby Multichannel Amplifier units to support more channels and can use the Dolby Multichannel Amplifier with the CP950/A or CP850 analog outputs and Dolby Atmos Interface DAC3202s (to provide audio to amplifiers with analog inputs). To connect more than three Dolby Multichannel Amplifiers and/or three DAC3202s, you must use a network switch instead of daisy chaining the amplifiers.

You cannot use the Dolby Multichannel Amplifier with a Dolby Atmos Interface DAC3201, as a different Dolby Atmos Connect protocol is used on that device.

The Dolby Multichannel Amplifier is available in several configurations. For part numbers and other related information, see Appendix D.

1.1 Dolby Multichannel Amplifier Front Panel

Following is a description of the Dolby Multichannel Amplifier front-panel components.

Figure 1-1 Dolby Multichannel Amplifier Front Panel

1.1.1 LED Channel and System Status

The front-panel LEDs specify whether audio is present on the amplifier outputs of the various channels. Blue LEDs illuminate faintly if no signal is present and increase in brightness that corresponds to the output signal level. For units with the CAT1416 analog-to-digital converter, the LEDs illuminate faintly in green if no analog signal is present and increase in brightness that corresponds to the analog input signal level. The LED indicators also alert you to various system states or if there is a problem with the amplifier.

Following are descriptions of the Dolby Multichannel Amplifier LED behavior for the various states:

- **Off mode**: None of the channel LEDs are illuminated. The power button illuminates in yellow.
- **Booting**: The signal LEDs begin illuminating in blue, from left to right (and will include green for units that include a CAT1416 analog-to-digital converter). When all the LEDs are lit, the system indicates that the booting process is complete by briefly illuminating the LEDs in white.
- **Channel clipping:** If the Dolby Multichannel Amplifier receives an excessive signal and is overdriven, the affected channel or channels may clip, which may be audible in the auditorium. Clipping is indicated when the signal LEDs illuminate in orange for the Digital Signal Processor (DSP) and full white for the output stage of the amplifier, for as long as the clipping occurs. Clipping indicators are also displayed in the **Status** screen of the web user interface. (For more information on the system status, see Section 3.2.1)
- **Channel shut down**: If there is a problem with the Dolby Multichannel Amplifier, wiring, or speakers, the unit may disable the affected channel or channels. The front-panel LEDs illuminate in red from the outer edges and meet in the center. After the LEDs illuminate in red, the affected channel LEDs blink red/yellow/red, and then the initial red LED sequence repeats. (For more information on the system status, see Section 3.2.1)
- **Shutdown mode due to hardware fault**: If the Dolby Multichannel Amplifier detects a critical problem and cannot compensate, all the red LEDs slowly illuminate on and off. In such a case, disconnect power from the unit and wait two minutes. Then power up the unit. You may need to check the **Status** screen for additional information regarding this fault (For more information on the system status, see Section 3.2.1)
- **Normal shutdown**: When the Dolby Multichannel Amplifier is shutting down to off mode, the blue LEDs slowly illuminate on and off. Once the system is in off mode, the only indication is the illuminated power button.
- **Power Standby Mode:** When the amplifier is in Standby Mode to save energy, the front panel power button will flash orange in coordination with the orange warning indicator in the web UI. Managing Power Standby Mode can be accomplished via the web UI and SNMP command.

1.1.2 Power Button

Press the power button to turn the Dolby Multichannel Amplifier on, and press and hold the power button for a few seconds to turn the unit off.

Following are descriptions of the power button LED behavior for the various states:

- **No illumination:** Disconnected mode. No AC power.
- **Solid yellow:** Off mode. System is plugged in but not powered on.
- **Flashing orange:** System is in Power Standby mode.

- **Flashing white:** System is booting (typically occurs after pressing power button).
- **Solid blue:** System has booted.
- **Flashing blue:** System is booting or shutting down.
- **Solid red:** A critical power or thermal fault has occurred. There is no audio output. If a

fault has occurred, momentarily pressing the button clears the fault.

1.1.3 USB Port

This port provides you with the capability of performing alternative maintenance procedures when you insert a USB drive, as described in Alternative Maintenance Procedures. It flashes in blue to indicate activity on this port. A solid USB icon appears at the top of the screen in the Web UI when a USB drive is connected. When data is being written, the icon will blink.

1.2 Dolby Multichannel Amplifier Rear Panel

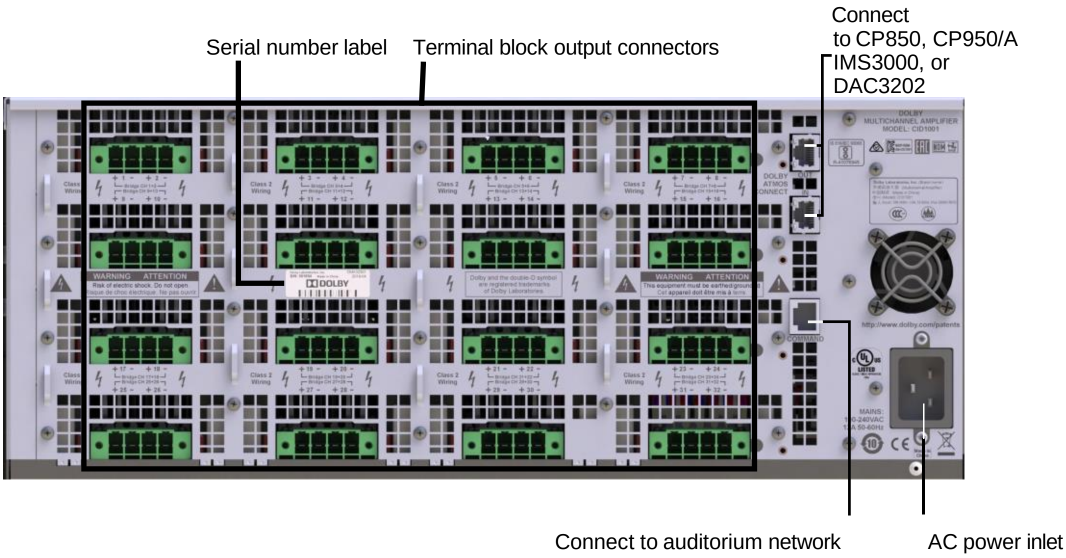

Figure 1-2 DMA32301 Rear Panel

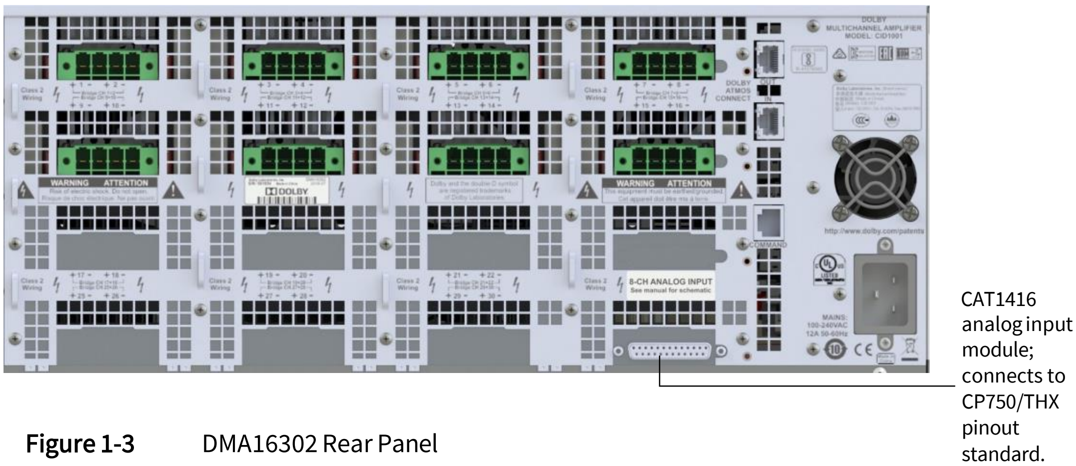

Figure 1-3 DMA16302 Rear Panel

Following is a description of the Dolby Multichannel Amplifier rear-panel components.

1.2.1 AC Mains Power Inlet

Connect the provided AC line cord (100 to 240 VAC) that is correct for your region to this port. This is an IEC 320-C20 20A AC power inlet.

1.2.2 Dolby Atmos Connect In/Out RJ-45 Ports

Use these ports to connect to the CP950/A, CP850, and IMS3000 (with an internal audio processor enabled), or DAC3202, as described in section 2.4 for single amplifier and section 2.5 for multiple amplifier installation.

1.2.3 Command (Com) R-J45 Port

Use this port to connect to the auditorium network switch. You perform all setups, control, and monitoring of a Dolby Multichannel Amplifier through this connection. The Dolby Multichannel Amplifier default IP address is 192.168.1.143.

1.2.4 Terminal Block Connectors

The Dolby Multichannel Amplifier has an array of two-channel terminal block connectors, depending upon the configuration. Each connector is labeled on the rear panel, which indicates its audio output channels, respective pinouts, and bridging information.

1.2.5 CAT1416 DB-25 Connector (DMA16302 and DMA24302 Only)

Use this 25-Pin D-connector to connect a Dolby CP750 (or other cinema processor with no compatible digital outputs) to input analog audio and convert it to digital audio for use by the Dolby Multichannel Amplifier.

Installing the Dolby Multichannel Amplifier

This chapter is for qualified installers and provides step-by-step instructions for installing the Dolby® Multichannel Amplifier. It covers:

- Rack-Mounting Instructions
- Connecting AC Power
- Connecting Audio Input
- Connecting a Single Dolby Multichannel Amplifier
- Connecting Two Dolby Multichannel Amplifiers
- Connecting a Dolby Multichannel Amplifier to a CP750
- Connecting the Output Channels to the Speakers

Depending on your region, you should have received the appropriate accessory kit for your Dolby Multichannel Amplifier:

- **DMA-ACC-US is for the United States and Canada.**
- **DMA-ACC-CN is for China.**
- **DMA-ACC-IN is for India**
- **DMA-ACC-KR is for Korea.**
- **DMA-ACC-TW is for Taiwan.**
- **DMA-ACC-ROW is for the rest of the world.**

These kits include the required AC line cord, terminal block output connectors (for the amplifier outputs), a printed quick start guide, cable management bars, and rack brackets (required for racks with pre-threaded or drilled holes).

2.1 Rack-Mounting Instructions

You need to rack mount Dolby Multichannel Amplifier units in a 4U rack space. You can order the optional quick-release rail kits for your amplifiers, depending on your requirements. These rails are designed for installation into racks with square holes.

- **CAT1240 is the short rail kit for racks that are 19 to 26 inches (48 to 66 cm) deep.**
- **CAT1140 is the long rail kit for racks that are 26 to 38 inches (66 to 97 cm) deep.**

The Dolby Multichannel Amplifier is 22 inches (56 cm) deep, but we recommend a rack that is at least 32 inches (81 cm) deep so that you can install the required cables and maintain proper airflow. In accordance with modern IT equipment conventions, the Dolby Multichannel Amplifier intakes air at the front panel and exhausts out the back of the unit.

You should inspect the Dolby Multichannel Amplifier and its shipping package and contact Dolby immediately if you find any damage.

Identify a suitable location for the rack unit that will hold the Dolby Multichannel Amplifier. It should be situated in a clean, dust-free area that is well ventilated. Avoid areas where heat, electrical noise, and electromagnetic fields are generated. You will also need to place it near a grounded power outlet. Read all the precautions listed in the next section.

2.1.1 Preparing for Setup

Leave at least 25 inches (63 cm) clearance in front of the rack and approximately 30 inches (76 cm) of clearance in the back of the rack to allow for sufficient air flow and access for servicing.

Rack Precautions

Be sure to take these precautions when installing the rack:

- Ensure that the leveling jacks on the bottom of the rack are fully extended to the floor with the full weight of the rack resting on them.
- In a single-rack installation, attach stabilizers to the rack.
- In multiple-rack installations, couple the racks together.
- Before extending a component from the rack, be sure the rack is stable.

Extend only one component in a rack at a time; extending two or more simultaneously may cause the rack to become unstable.

General Component Precautions

Be sure to take these precautions when installing all components in the rack:

- Review the electrical and general safety precautions.
- Identify the placement of each component in the rack before you install the rails.
- Install the heaviest rack components on the bottom of the rack first, and then work up to the lightest.
- Close all panels, all components, and rack door (if present) when not servicing, to maintain proper cooling.

Rack-Mounting Considerations

Be sure to consider the following when installing the rack.

Ambient Operating Temperature

If installed in a closed or multiunit rack assembly, the ambient operating temperature of the rack environment may be greater than the ambient temperature of the room. Therefore, consideration should be given to installing the equipment in an environment compatible with the manufacturer’s maximum rated ambient temperature.

Airflow

Equipment should be mounted into a rack so that the amount of airflow required for safe operation is not compromised. The Dolby Multichannel Amplifier airflow is from front to back.

Mechanical Loading

Equipment should be mounted into a rack so that a hazardous condition does not arise due to uneven equipment loading.

Reliable Ground

A reliable ground must be maintained at all times. To ensure this, the rack itself should be grounded. Particular attention should be given to power supply connections other than the direct connections to the branch circuit (such as the use of power strips).

Circuit Overloading

You must connect the Dolby Multichannel Amplifier to a dedicated circuit. Be sure to consider the connection of the equipment to the power supply circuitry and the effect that any possible overloading of circuits might have on over-current protection and power supply wiring. Appropriate consideration of equipment name-plate ratings should be used when addressing this concern.

2.1.2 Installing the Dolby Multichannel Amplifier in a Rack Using the Long or Short Rails

Following are instructions for installing the Dolby Multichannel Amplifier into its rack using the long or short rails. There are several types of racks available, which may require slight variations in the installation procedure. You should also refer to the installation instructions provided with your rack.

Note: If your rack does not have square holes, you will need to install the adapter bracket that is included in your accessory kit (along with the AC line cord).

Rail kits are shipped separately from the accessory kits. There are two rail assemblies provided for the rails. Each assembly consists of two sections, an inner fixed chassis rail that secures directly to the Dolby Multichannel Amplifier and an outer fixed rack rail that secures directly to the rack itself.

To install the optional quick-release rails:

1. Separate the inner and outer rails on each assembly (refer to Figure 2-1):
    - Extend the rail assembly by pulling it outward.
    - Press the quick-release tab.
    - Separate the inner rail extension from the outer rail assembly.

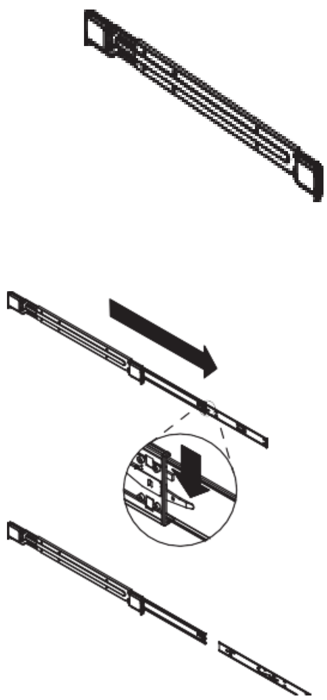

Figure 2-1 Separating the Small Quick-Release Rails

2. Install the inner rail extensions, as shown in Figure 2-2:
   - Place the inner rack extensions on each side of the Dolby Multichannel Amplifier chassis, aligning the hooks of the Dolby Multichannel Amplifier chassis with the rail extension holes.
   - Be sure each extension faces outward just like the inner rail.
   - Slide the first extension toward the front of the Dolby Multichannel Amplifier chassis.
   - Secure the inner rail to the Dolby Multichannel Amplifier chassis with the two provided M4 flat-head screws, as shown in Figure 2-2. Repeat these steps for the other inner rail.

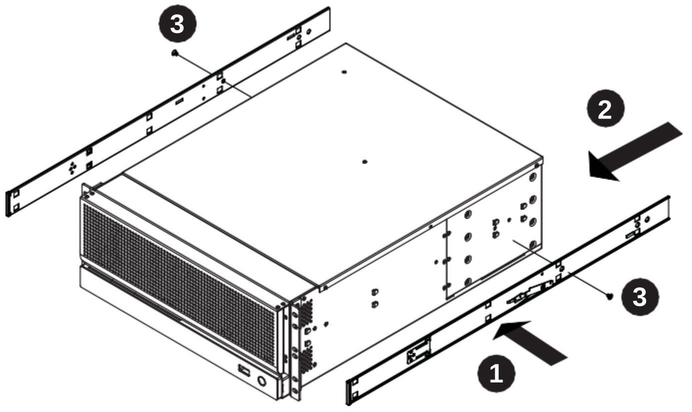

Figure 2-2 Installing the Inner Rail Extensions on the Small Rails

3. Assemble the outer rack rails, which attach to the rack:
   - Secure the back end of the outer rail to the rack, using the provided screws.
   - To retract the smaller outer rail, press the button where the two outer rails join.
   - Hang the rail hooks on the rack holes, and if desired, use the provided screws to secure the front of the outer rail to the rack.
   - Repeat steps 1–3 for the other outer rail.

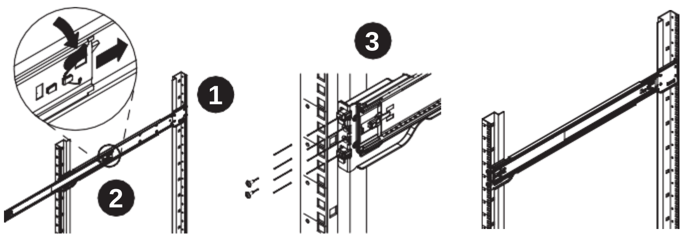

Figure 2-3 Assembling the Outer Small Rails

4. Install the Dolby Multichannel Amplifier into the rack, as shown in the following figure:
   - Confirm that the inner and outer rails are installed on the rack.
   - Extend the outer rails.
   - Align the inner rails on the Dolby Multichannel Amplifier with the outer rails on the rack.
   - Slide the inner rails into the outer rails, maintaining equal pressure on both sides. When the Dolby Multichannel Amplifier is completely inserted into the rack, it should click into the locked position.

- Insert and tighten the 10-32 rack screws in the center hole locations to secure the Dolby Multichannel Amplifier to the rack.
- Rail brackets (included with the regional accessory kits) are required for racks with pre-threaded or drilled holes.

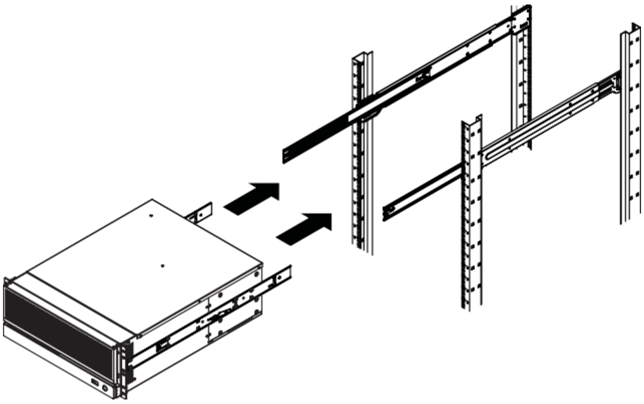

Figure 2-4 Installing the Dolby Multichannel Amplifier in the Rack Using the Small Rails

5. Connect all cables and configure the system, as described in the following sections.

Two cable management bars (Part Number 6201910) are shipped with the regional accessory kits. These bars are slotted for cable ties and attach to the rear side of your rack to support the weight of the cables. We recommend that you minimize any tension on the terminal block connectors when using large-gauge cables.

Figure 2-5 Cable Management Bars

2.1.3 Additional Requirements

The Dolby Multichannel Amplifier vents from front to back. Be sure to provide proper clearance and unobstructed airflow. Do not install the units above heat-generating equipment. For CP750, CP850, and CP950/A configurations, the Dolby Multichannel Amplifier should be mounted in the same rack as the CP750, CP850, or CP950/A, to avoid potential problems with ground loops and radiated interference.

Note: Follow all local codes and regulations covering electrical wiring.

2.2 Connecting AC Power

Following are the Dolby Multichannel Amplifier AC input requirements:

- AC voltage: 100 to 240 VAC
- Type of connection: Dedicated breaker circuit

Breaker current: The Dolby Multichannel Amplifier supports a 15-, 16- or 20-amp circuit. The web client default configuration is 15 amps. You must configure this parameter in the Dolby Multichannel Amplifier web client if a 16- or 20- amp breaker is used; If the circuit is left at 15-amp but the circuit is 16- or 20-amp, the unit limits prematurely and does not provide the maximum amplification.

2.3 Connecting Audio Input

You can connect a Dolby Multichannel Amplifier in different configurations, depending upon your requirements. Each Dolby Multichannel Amplifier receives digital audio and outputs amplified audio based on the number and type of amplifier cards that are installed in the unit.

Note: If more channels are required, you can add additional units for amplified channels or Dolby DAC3202s for analog (nonamplified) channels. In either case, you cannot combine more than three of these devices unless you use a switch. For more information on switch configurations, see Section 2.5.1.

Note: You cannot combine the Dolby Multichannel Amplifier or DAC3202 outputs with a Dolby Atmos® Interface DAC3201. The DAC3201 uses a different protocol that is not supported by the Dolby Multichannel Amplifier or DAC3202.

The following figure shows some of the different configurations for interconnecting Dolby audio processors that provide compatible digital outputs. For audio processors with no compatible digital outputs, see Section 2.6 for connecting to a DMA16302 or DMA24302. For all other DMA units, to interconnect with audio processors with no compatible digital outputs, see the separate *Dolby CP750 7.1 audio for the Dolby Multichannel Amplifier Installation Manual.*

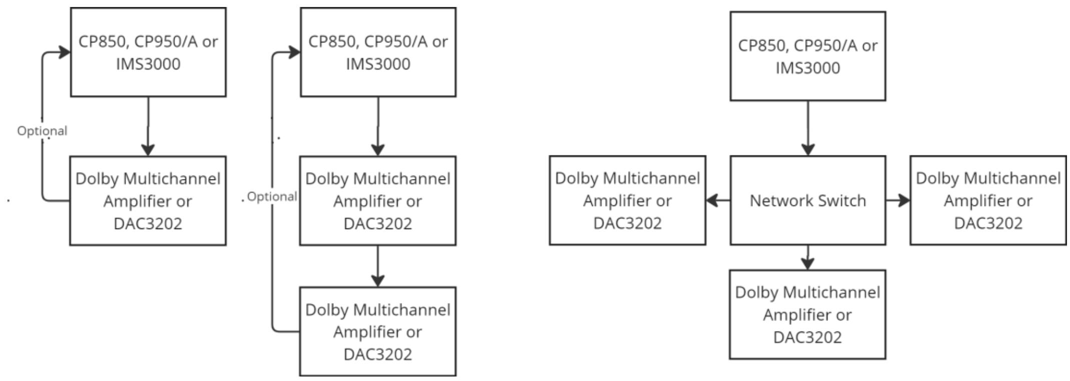

Figure 2-6 Dolby Multichannel Amplifier Configurations Using Dolby Atmos Connect

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Note: If your installation still requires more channels, you can also utilize the CP950/A or CP850 analog outputs, or the IMS3000 digital outputs.

2.4 Connecting a Single Dolby Multichannel Amplifier

To connect a Dolby Cinema Audio Processor (CP850, CP950/A, or IMS3000) to a single Dolby Multichannel Amplifier:

1. Use a shielded Cat5e or greater cable to connect the Dolby Multichannel Amplifier **COMMAND** Ethernet port to the auditorium network switch.
2. Use a second shielded Cat5e or greater cable to connect the Cinema Processor **DOLBY ATMOS CONNECT OUT** port to the **DOLBY ATMOS CONNECT IN** port on the Dolby Multichannel Amplifier rear panel.
3. Use a third shielded Cat5e or greater cable to connect the CP850, IMS3000, or CP950/A **DOLBY ATMOS CONNECT IN** port to the **DOLBY ATMOS CONNECT OUT** port on the Dolby Multichannel Amplifier. Note that this cable is optional.

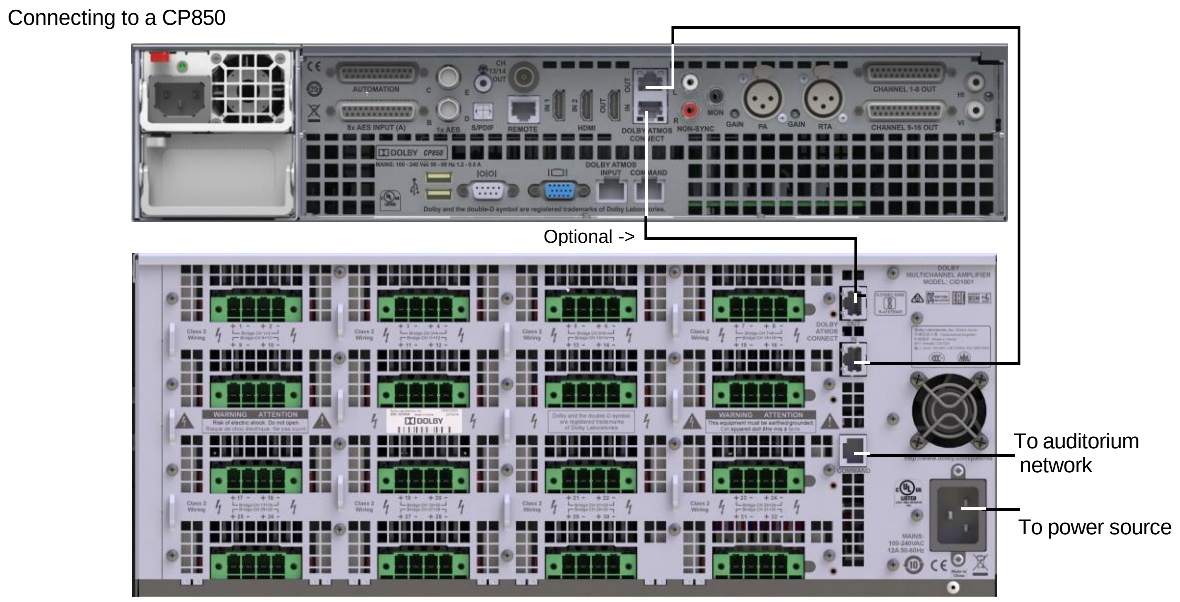

Figure 2-7 Connecting a Single Dolby Multichannel Amplifier to a CP850

Connecting to an IMS3000

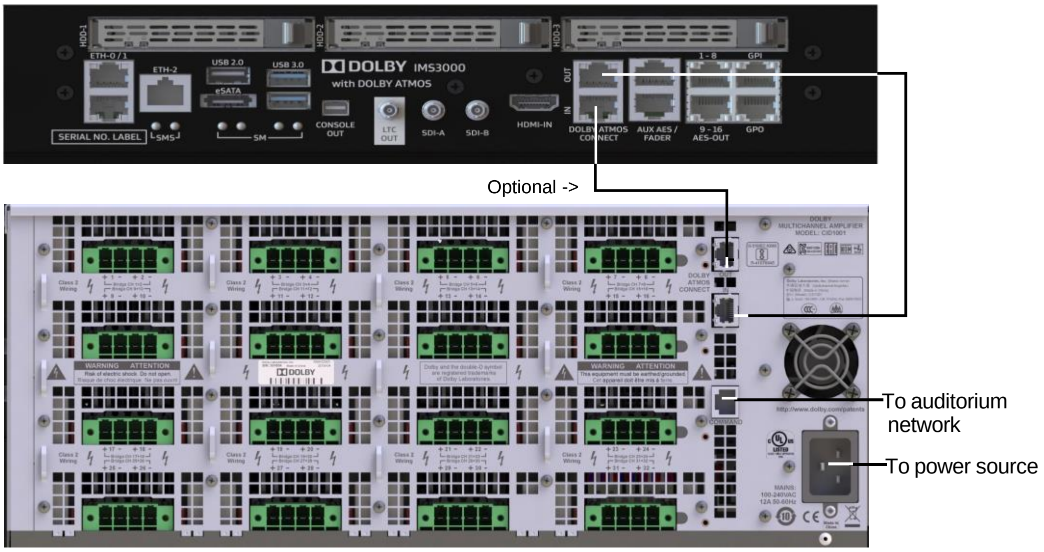

Figure 2-8 Connecting a Single Dolby Multichannel Amplifier to an IMS3000

Connecting to a CP950

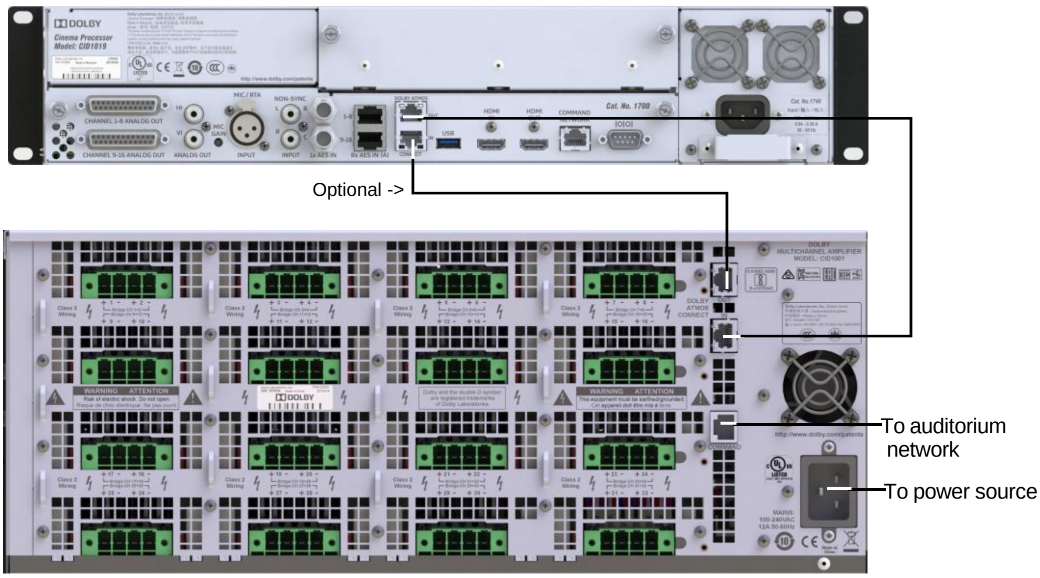

Figure 2-9 Connecting a Single Dolby Multichannel Amplifier to a CP950 (CP950A uses the same ports)

4. Plug the unit into any dedicated electrical circuit with a line input voltage from 100–240 VAC, press the power button to boot up the unit, and then connect your computer to the auditorium switch.

Figure 2-10 Dolby Multichannel Amplifier Front Panel

5. Configure your computer IP address to communicate with the Dolby Multichannel Amplifier default IP address (192.168.1.143) on the **COMMAND** port. We recommend that you enter an IP address above 192.168.1.150 and below 192.168.1.254, with a subnet mask of 255.255.255.0.
6. Open your web browser and enter the Dolby Multichannel Amplifier **COMMAND** port default IP address 192.168.1.143 to display the web client user control screen. (Currently, we recommend the Google™ Chrome™ browser.)
7. Under the **Command** section of the **Network** tab, click **Manual** for **IP configuration**. We recommend that you change the **IP address**, **Netmask**, and **Gateway** settings to work with your IP schema by clicking in the respective fields, making the required entries, and clicking **Apply**. When prompted for credentials, the default password is *password*.

If you are using the Dolby default IP scheme, we recommend that you change the third octet of the default IP address to match your auditorium number and the fourth octet to 143. This will be 192.168.x.143, where x = auditorium number.

8. Proceed to chapter 3.

Note: The Dolby Multichannel Amplifier boots up in approximately 3.5 minutes.

During start-up, the power button on the Dolby Multichannel Amplifier front panel flashes on and off in white and the signal indicators illuminate in blue from left to right. When the boot process is complete, the power button and the LED signal indicators (for each available channel) illuminate in blue. For CAT1416 CP750 input channels, the signal LEDs illuminate in green.

2.5 Connecting Two Dolby Multichannel Amplifiers

To connect a Dolby Cinema Processor (CP850, CP950/A, or IMS3000) to two Dolby Multichannel Amplifier units:

1. Connect the **COMMAND** port on each Dolby Multichannel Amplifier to the auditorium network switch.
2. Connect the Cinema Audio Processor **DOLBY ATMOS CONNECT OUT** port to the **DOLBY ATMOS CONNECT IN** port on the first Dolby Multichannel Amplifier rear panel.
3. Connect the **DOLBY ATMOS CONNECT OUT** port on the first Dolby Multichannel Amplifier to the **DOLBY ATMOS CONNECT IN** port on the next Dolby Multichannel Amplifier.
4. If another Dolby Multichannel Amplifier is used, follow step 3 again with the next unit.
5. Connect the last Dolby Multichannel amplifier **DOLBY ATMOS CONNECT OUT** port to the **DOLBY ATMOS CONNECT IN** port on the Dolby Cinema Processor.

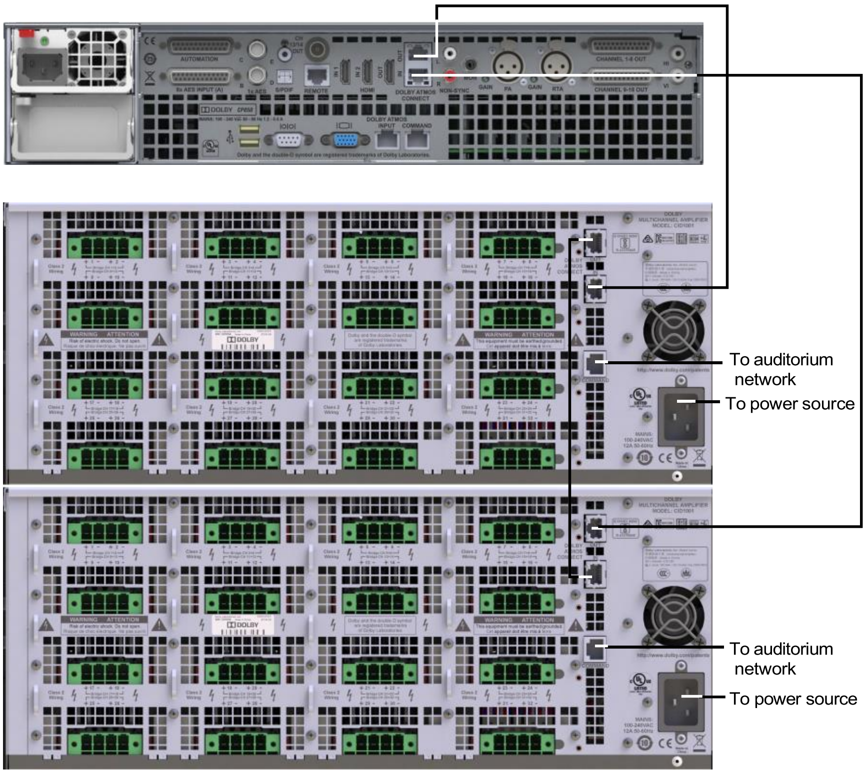

Figure 2-11 Connecting Two Dolby Multichannel Amplifiers to a CP850

Note: Be sure to use shielded Cat5e or greater cables for the input/output connections.

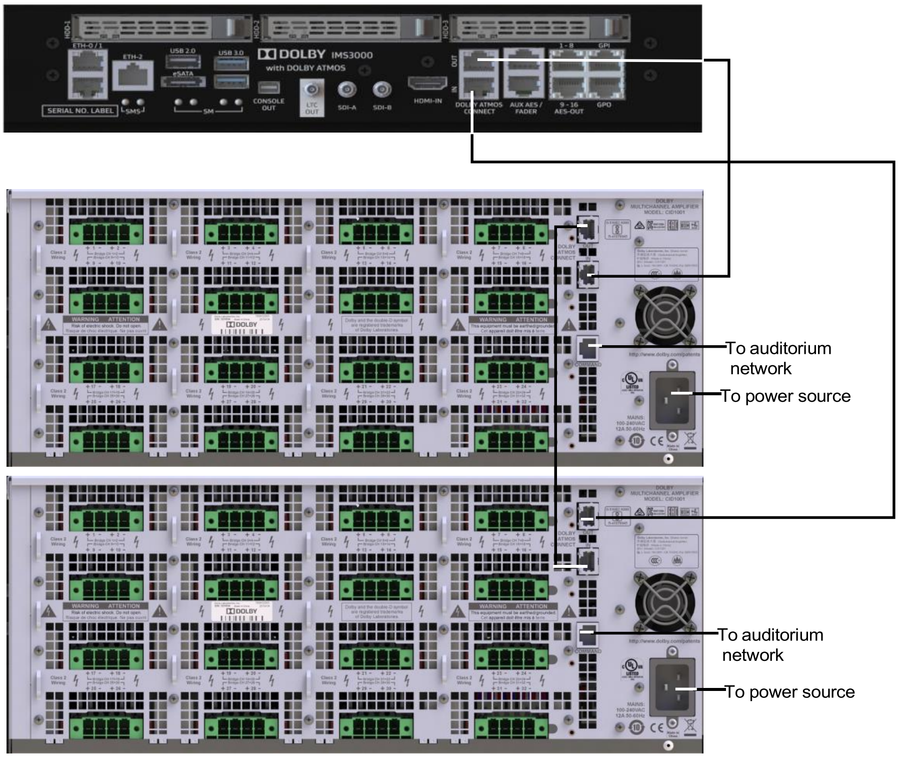

Figure 2-12 Connecting Two Dolby Multichannel Amplifiers to an IMS3000

Connecting to a CP950 (shown) or CP950A

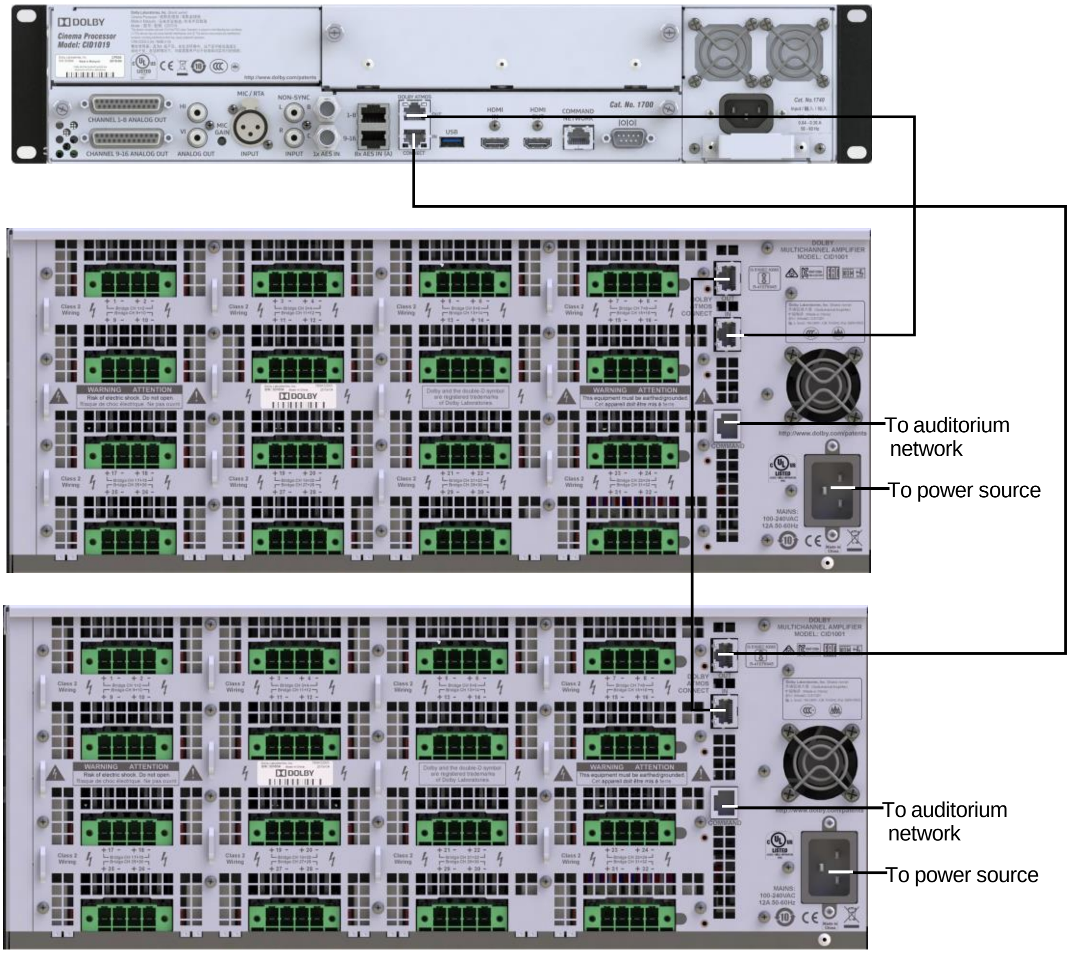

Figure 2-13 Connecting Two Dolby Multichannel Amplifiers to a CP950

6. Refer to steps 5 through 7 in Section 2.4 for connecting a single unit, and apply these to the second unit.
7. Plug the second Dolby Multichannel Amplifier into a dedicated electrical circuit with a line input voltage from 100 to 240 VAC. If you are using the Dolby default IP scheme, we recommend that you change the fourth octet of the second unit to 144, which results in an IP address of 192.168.x.144 for the second channel group, where x = auditorium number. (The previously configured IP address for the first unit is 192.168.x.143 for the first channel group.)
8. If a third Dolby Multichannel Amplifier is required, and you are using the Dolby default IP scheme, we recommend that you change the fourth octet of the second unit to 145, which results in an IP address of 192.168.x.145 for the third channel group, where x = auditorium number. (The previously configured IP address for the second unit is 192.168.x.144 for the second channel group.)

Note: If more channels are required, you can add additional units for amplified channels or Dolby DAC3202s for analog (nonamplified) channels. In either case, you cannot interconnect more than three of these devices unless you use a switch. For detailed information on using a switch, see Section 2.5.1.

Note: Be sure to use shielded Cat5e or greater cables for the input/output connections.

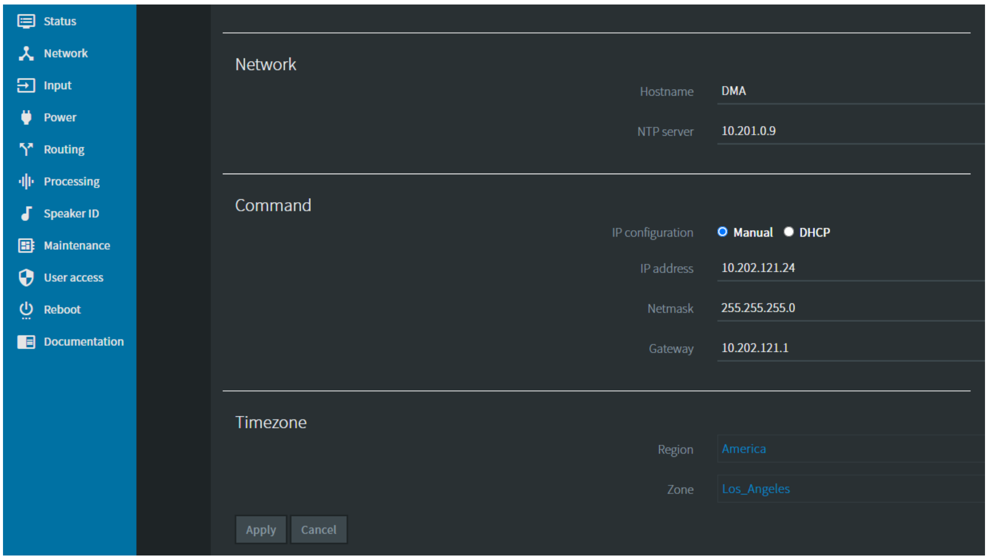

Figure 2-14 Change Default IP Scheme

9. Click **Input** in the navigation menu to select an **Input Type** = **digital** for the CP950/A, CP850, and IMS3000, or **analog** for the CP750.

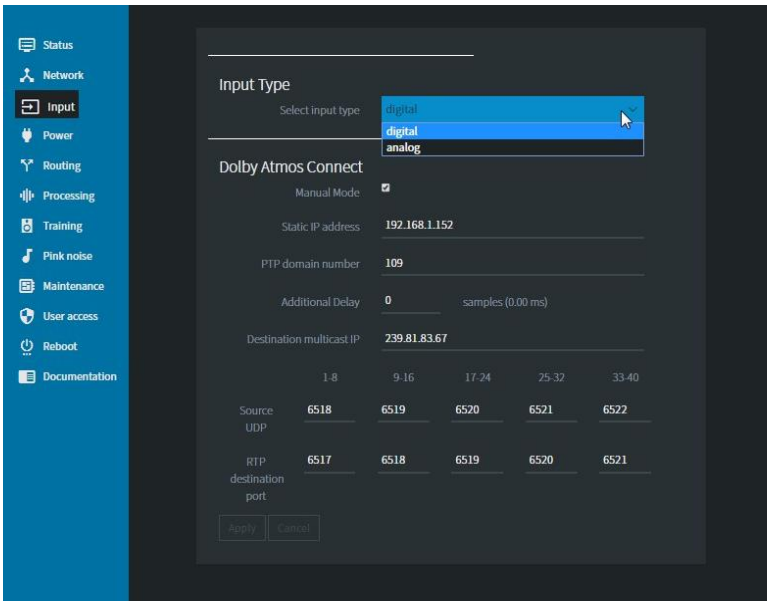

Figure 2-15 Select an Input Type

2.5.1 Using Dolby Multichannel Amplifiers or DAC3202s with a Switch

In situations where three or more devices (Dolby Multichannel Amplifier, or DAC3202) are needed for an installation, Dolby recommends routing the audio through an Ethernet network switch. This ensures that each of the units receives the same timing messages to keep the audio synchronized.

1. Connect the **Dolby Atmos Connect OUT** port from the Dolby CP950/A, CP850 or IMS3000 to the Ethernet network switch.
2. Connect the **Dolby Atmos Connect IN** ports on the Dolby Multichannel Amplifiers and/or Dolby DAC3202s to the same switch. No loop-back cabling is required or recommended.

Following are the desired switch specifications for interconnecting four or more devices:

- For a dedicated network, we recommend a switch that provides Gigabit speed to create a star-topology audio network.
- Dolby recommends disabling IGMP snooping on network switches.
- For a non-dedicated network, we recommend a managed switch that supports:
    - Gigabit speed
    - DiffServ Quality of Service (QoS) with strict priority and four queues

2.6 Connecting a Dolby Multichannel Amplifier to a CP750

A Dolby CP750 cinema processor (or another cinema processor) that has analog outputs (but no Dolby Atmos Connect digital outputs) can be connected to a Dolby Multichannel Amplifier that includes a CAT1416 analog-to-digital-converter card. Currently, DMA units with a CAT1416 are the DMA16302 and DMA24302. Earlier DMA units can receive digital audio from a CP750 or other audio processors via a Yamaha Tio1608-D analog-to-digital converter. For information on this type of configuration, log into https://customer.dolby.com/cinema/ and download the *Dolby CP750 7.1 audio for the Dolby Multichannel Amplifier Installation Manual*.

1. Connect the CP750 **MAIN AUDIO OUTPUT** (THX pinout) to the amplifier CAT1416 audio input using a male-to-female 25-Pin D-Connector cable.

You can use the optional Dolby cable (Part Number DMA-ACC-ANA-CBL), which has shielded audio pairs to limit noise and crosstalk, or you can use your own cable that meets these specifications (For wiring information, see CAT1416 Cable Pinouts.)

The following figure shows a CP750 audio connection to a DMA16302.

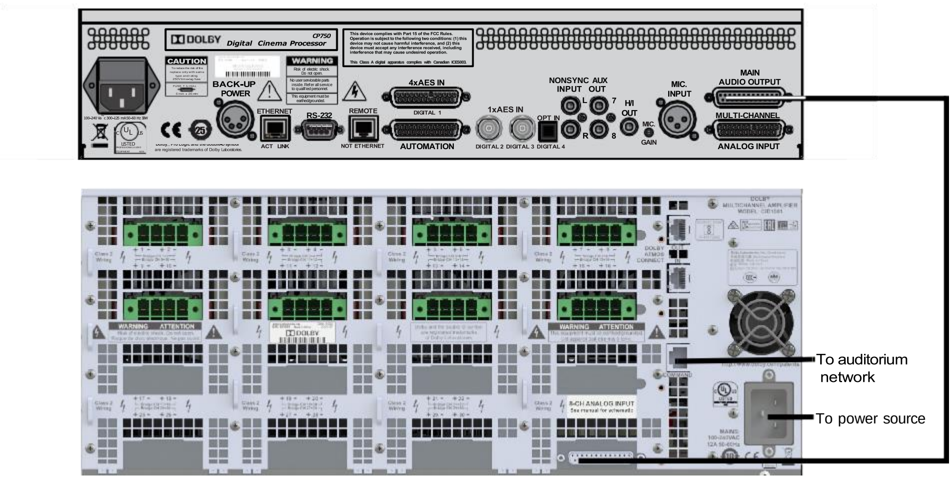

Figure 2-16 Connecting to a CP750

When connected to the Dolby Multichannel Amplifier inputs, the last LED signal indicators will illuminate in solid green (in the web UI the indicators will be listed as “Inputs”. When a CAT1416 analog-to-digital converted 7.1 or 5.1 signal is present, these LEDs flash in green. When a CAT1416 output signal is present, these LEDs illuminate in blue.

2. Click **Input** in the left navigation menu then select the **analog** Input Type.

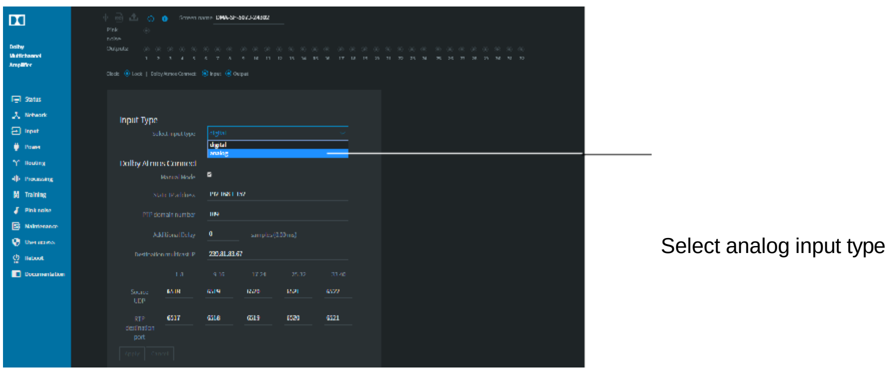

Figure 2-17 Select an Analog Input

2.7 Connecting the Output Channels to the Speakers

The Dolby Multichannel Amplifier supports output channels in both normal and bridge mode.

In normal mode:

- For units with the CAT1433 high-power amplifier cards, each channel can provide up to 300 watts of audio power into an 8-ohm speaker, up to 600 watts into a 4-ohm speaker, and up to 600 watts into a 2-ohm speaker. The DMA16301, DMA16302, DMA24302, and DMA32301 include the CAT1433 card.
- For units with the CAT1422 lower-powered amplifier cards, each channel can provide up to 300 watts of audio power into an 8-ohm speaker or 4-ohm speaker. The DMA16300, DMA24300, and DMA32300 include the CAT1422 card.

In bridge mode:

- For units with the CAT1433 high-power amplifier cards, you can bridge a channel pair to supply up to 1100 watts into an 8 ohm or 4-ohm speaker. Bridging on these cards is not supported on 2-ohm speakers.
- For units with the CAT1422 lower-powered amplifier cards, you can bridge a channel pair to supply up to 600 watts. Bridging on these cards is supported only on 8-ohm speakers.

Note: For systems with version 1.0.x.x software, bridging channels results in a loss of the even numbered output for use on other speakers. For example, when you bridge channels 1 and 2, you cannot use channel 2. Upgrading to version 2.0.x.x software or higher provides improved routing options that allow you to use the even numbered channels.

To determine the number and type of amplifier cards installed in a Dolby Multichannel Amplifier, click the system info button at the top of the web client user-control screen.

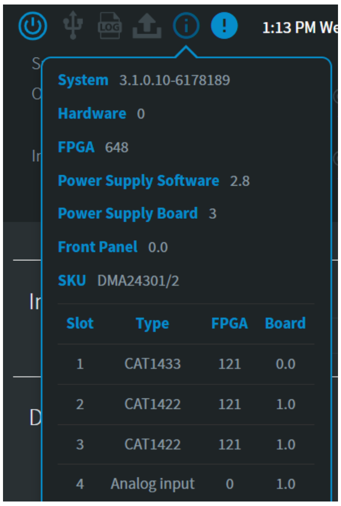

Figure 2-18 Click Info Button to Display System Information

The Dolby Multichannel Amplifier outputs are arranged in terminal blocks. In normal mode, each Dolby Multichannel Amplifier terminal block supports two loudspeaker terminations. In bridge mode, each terminal block supports a single loudspeaker termination. In either mode, you connect a speaker using two wires. One wire is positive (+), and the other wire is negative (–).

The speaker wires interface to the Dolby Multichannel Amplifier using a mating four-way terminal block plug, as shown in the following figures. Sixteen of these plugs are included in your accessory kit. You secure these plugs to each side of the terminal blocks with the provided screws. These plugs can accept up to 8 AWG stranded speaker wire with insulation stripped back 6-7 mm. When connected, there should be no bare wires exposed on the connector as this could be a shock hazard. You should secure the wires into the connector with a torque rating of approximately 7 in-lb.

Warning: Disconnect the AC mains power prior to making any connections. When the AC power is on, there may be dangerous voltage at the terminal blocks on the Dolby Multichannel Amplifier rear panel. Do not touch these contacts.

Be sure to use Class-2 wiring for connections.

2.7.1 Connecting Channels in Normal Mode

To connect a channel pair on a terminal block to a speaker in normal mode:

1. Locate the two connectors on the left of the terminal block plug.
2. Connect the positive (+) output of the amplifier to the positive (+) input terminal of a speaker.
3. Connect the negative (–) output of the amplifier to the negative (–) input terminal on the same speaker. (See the figure below)

To connect a second channel pair on the terminal block to a speaker in normal mode:

1. Locate the two connectors on the right side of the terminal block plug.
2. Connect the positive (+) output of the amplifier to the positive (+) input terminal of a speaker.
3. Connect the negative (–) output of the amplifier to the negative (–) input terminal on the same speaker.

Normal mode

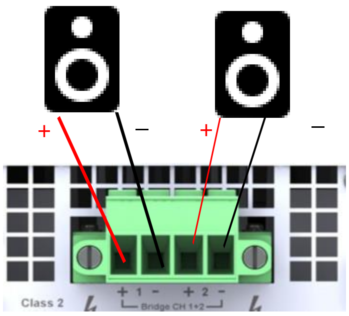

Figure 2-19 Connecting Channels from the Terminal Block to Speakers in Normal Mode

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Note: Be sure to use appropriate wire gauge, based upon loudspeaker impedance and the distance between the Dolby Multichannel Amplifier and the loudspeaker. The rear panel output terminal blocks are designed to accept up to 8-gauge wire.

2.7.2 Connecting Channels in Bridge Mode

Note: For earlier Dolby Multichannel Amplifiers with version 1.x.x software installed, bridging channels results in a loss of the even numbered output for use on other speakers. For example, when you bridge channels 1 and 2, you cannot use channel 2.

Upgrading to version 2.x.x software or higher provides improved routing options that allow you to use the even numbered channels.

Warning: Disconnect the AC mains power prior to making any connections. When the AC power is on, there may be dangerous voltage at the terminal blocks on the Dolby Multichannel Amplifier rear panel. Do not touch these contacts. Be sure to use Class-2 wiring for your connections.

To connect two channels in bridge mode, use the connectors that are furthest left and furthest right:

1. Connect the positive (+) output on the first channel (left terminal of the terminal block plug) to the positive (+) input on a speaker.
2. Connect the negative (-) output on the second channel (right terminal of the terminal block plug) to the negative (-) input on the same speaker.

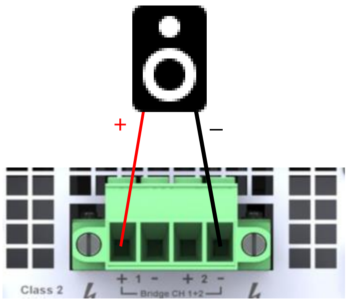


Figure 2-20 Connecting Channels from the Terminal Block to a Speaker in Bridge Mode

For bridge mode, you must configure the channel pair using the web client. (See Section 3.2.6)

Note: For the high-powered CAT1433 amplifier cards, when configuring the Dolby Multichannel Amplifier in bridged mode, be sure that the rated loudspeaker impedance is no less than 4 ohms. For the lower-powered CAT1422 cards, in bridged mode, be sure that the rated loudspeaker impedance is no less than 8 ohms.

Note: Be sure to use appropriate wire gauge, based upon loudspeaker impedance and the distance between the Dolby Multichannel Amplifier and the loudspeaker.

The rear-panel output terminal blocks are designed to accept up to 8-gauge wire.

Using the Dolby Multichannel Amplifier

After installing one or more Dolby® Multichannel Amplifiers in your auditorium network, as described in Chapter 2, you can configure and operate your system, as described in this chapter. This chapter covers the following:

- Connecting to the Dolby Multichannel Amplifier
- Navigation Bar
  - Status
  - Network
  - Input
  - Power
  - Routing
  - Processing
  - Speaker ID
  - Maintenance
  - User Access
  - Reboot
  - Documentation

3.1 Connecting to the Dolby Multichannel Amplifier

To connect to a Dolby Multichannel Amplifier, open your web browser, and enter the Dolby Multichannel Amplifier **Command** port IP address to display the web client user-control screen. The default IP address is 192.168.1.143. (Currently, we recommend only the Google™ Chrome™ or Firefox browser.)

The web-client user-control screen appears, and the **Status** tab is active, as indicated by its black highlighting in the navigation bar at the left side of the screen (Chrome only).

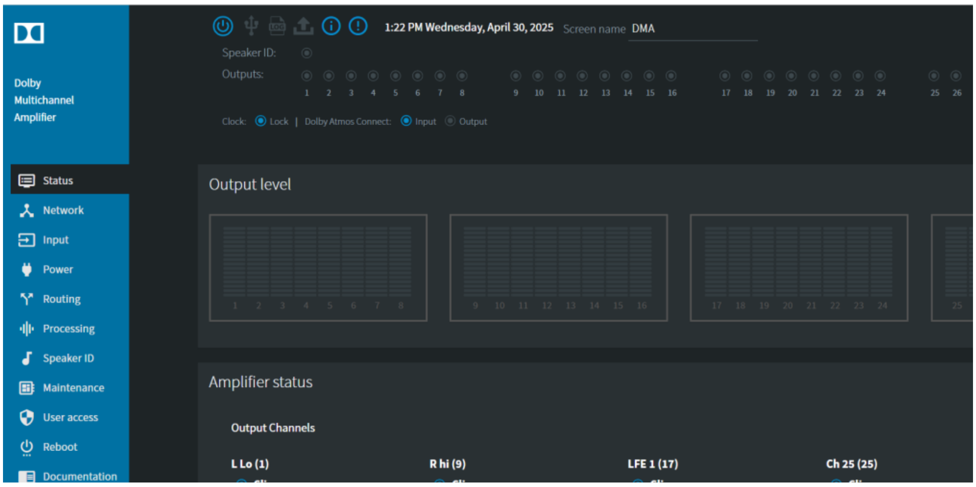


Figure 3-1 Status Screen

Across the top of the UI, there are several indicators that provide information or allow control of the device.

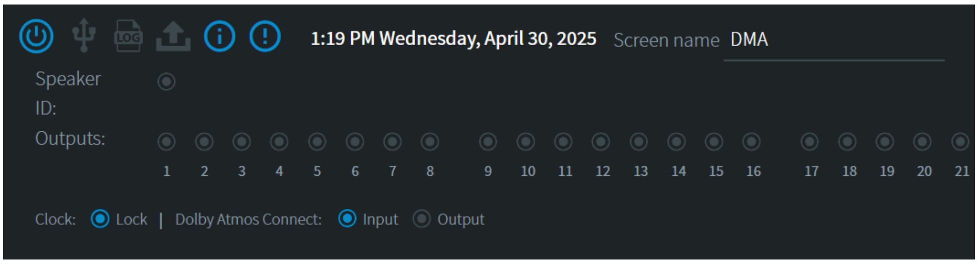


Figure 3-2 Status Screen Top Section

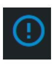
Power Icon: This icon shows the current state of the amplifier. Clicking on it allows you to put the system into low power standby mode or wake it from that state.

USB Icon: Indicates activity on the USB port.

Log Icon: Blinks when downloading logs.

Upload Icon: Blinks when uploading an upgrade package.

Info Icon: Click here to view system information.



Exclamation/Alert Icon: Click here to review reported system issues.

Date and time: will be displayed if the Dolby Multichannel Amplifier is connected to a valid NTP (Network Time Protocol) time server.

Screen name: A user can enter/edit a “Screen name” for the amplifier. This is a name that is shown in the UI and the exported log filename.

Speaker ID: Lights up when there is Speaker ID noise being played out of the amplifier.

Outputs: This is a graphical representation of the output stage of the amplifier.

Clock: When receiving a digital audio stream from a Dolby Cinema Processor (or other device) the **Lock** indicator will light up in blue when a valid digital audio clock is detected.

Dolby Atmos Connect port LEDs illuminate in blue if an Ethernet cable is connected to a Dolby Cinema Processor, another DMA, or other device and the **Dolby Atmos Connect** parameters are properly configured. (See Section 3.2.3.) These areas of the web client, along with the navigation bar, appear in all Dolby Multichannel Amplifier screens, which are described in the following sections.

For additional details in the **Status** screen, see Section 3.2.1.

3.2 Navigation Bar

The Dolby Multichannel Amplifier navigation bar provides access to the Dolby Multichannel Amplifier **Status**, **Network**, **Input**, **Power**, **Routing**, **Processing**, **Speaker ID, Maintenance**, and **User access** parameters. In addition, you can **reboot** the system and access user **documentation** from the navigation bar. Click the desired menu option to display the corresponding screen. To reconfigure most of these parameters and perform other functions, you need to enter your password at the prompt. (The default password is `password`.) For most installation workflows, you start at the top of the navigation bar and work your way down through the different parameters.

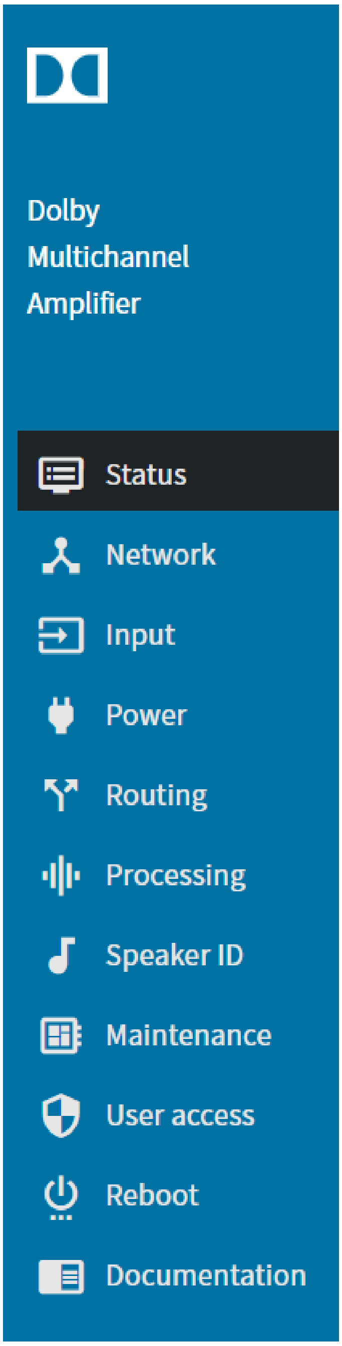


Figure 3-3 Navigation Bar

3.2.1 Status

When you click **Status** in the navigation bar, you can check the **Output Levels**, **Amplifier**, **Fans**, **Temperature**, and **Power Supply** status, as indicated by the respective blue (normal), yellow (warning) or red (fault) illumination. Note that certain conditions that cause a fault may require the amp to be rebooted to clear the fault signal, as it is assumed that these situations are serious, such as a blown speaker or short circuit, and the amplifier would need to be powered down to address them.

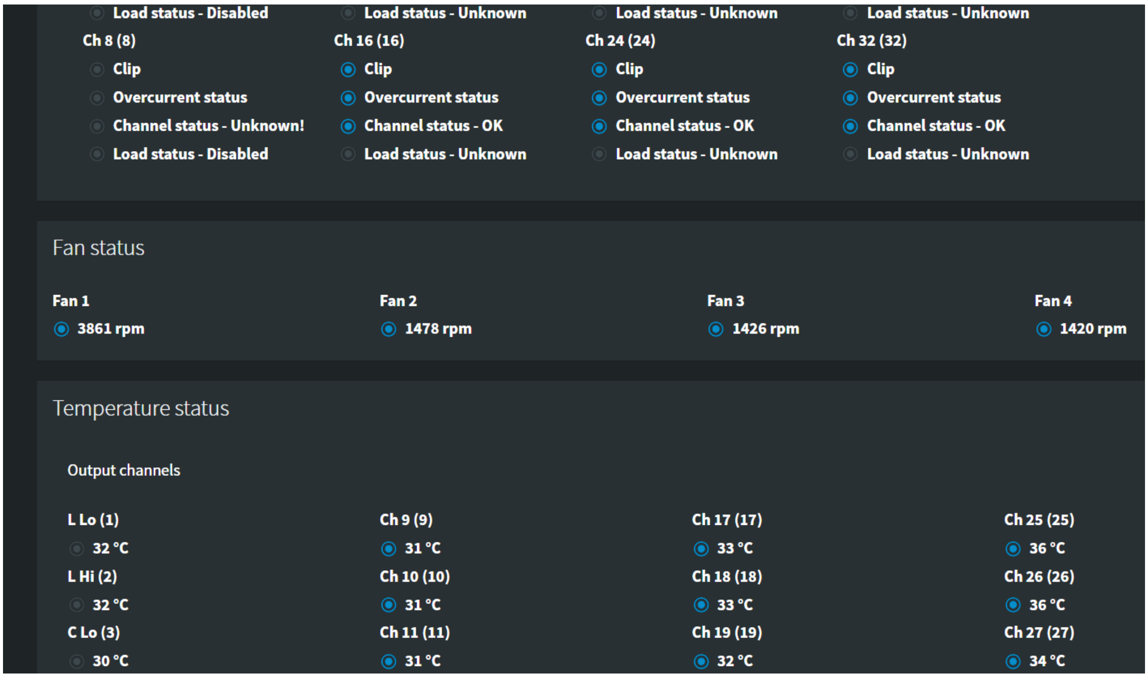


Figure 3-4 Status Screen

The following figure shows the eight analog input channels (in the red box) for a unit with a CAT1416 installed.

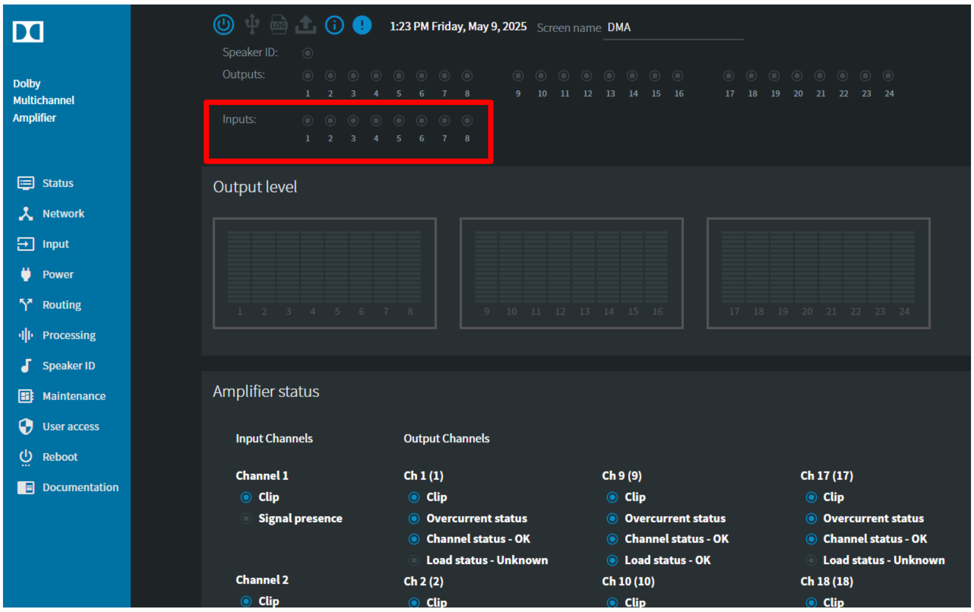


Figure 3-5 Status Screen with CAT1416 Input Channels

3.2.2 Indicators

Throughout the web UI there are indicators for level, clipping and general amplifier activity. Below are the descriptions for these various indicators (refer to the previous figure):

TOP STATUS SECTION

Digital output

- Grey - no signal
- Blue - signal present
- Red - silence detection fail, load fault, open relay, hi temp shutdown
- White - output clip
- Orange - digital clip
- Yellow - hi temp warning

Analog Input

- Grey - no signal
- Blue - signal present
- Red - signal clip

AMPLIFIER STATUS SECTION

Clip

- Grey - status unknown
- Blue - status ok
- Red - output clip

Overcurrent

- Grey - status unknown
- Blue - status ok
- Red - current limit error

Channel Status

- Grey - status unknown
- Blue - status ok
- Red - silence detection fail, open relay, hi temp shutdown
- Yellow - hi temp warning

Load Status

- Grey - status unknown or “disabled” when monitoring is not turned on
- Blue - status ok
- Red - speaker short or speaker open

Analog Channel Input Status CLIP indicator

- Grey - status unknown
- Blue - status ok
- Red - input clip

Analog Signal Presence

- Grey - no signal
- Blue - signal present

FAN STATUS SECTION

Fan Status indicators

- Grey – unknown status
- Blue – fan OK
- Orange – fan warning
- Red – fan fault

TEMPERATURE STATUS SECTION

Output Channels Temperature Status indicators

- Grey – not monitoring
- Blue – temp ok
- Yellow – hi temp warning
- Red – hi temp shutdown

Power Supply Temperature indicators:

- Blue - temp ok
- Red - hi temp error
- Power Supply Status LEDs
- Grey - status unknown
- Blue - status ok
- Red - fault

POWER SUPPLY STATUS SECTION

- Internal A/C line voltage – the current voltage received by the system
- Current draw – the amount of current drawn by the system at that moment
- Power status – Reports issues with the main power supply for the amplifier
    - Grey – Status unknown
    - Blue – Status OK
    - Red – Fault
- CAT1426 (PFC) fuse status – indicates if any of the fuses on the Power Factor Corrector board are not working.
- CAT1423 (LLC) fuse status – indicates if any of the fuses on the CAT1423 (inductor inductor capacitor) board are not working.

3.2.3 Network

To configure the network parameters, click **Network** in the navigation bar. The **Network** screen appears. In this screen, you can enter the hostname and the Network Time Protocol (NTP) server settings. With a valid NTP connection, the amplifier will display current date/time at the top of the screen.

When using NTP, set the Time Zone to correctly show the local time. This useful in configuring the power standby schedule.

You can also configure the **COMMAND** port settings, as described in Section 3.2.3.

To reconfigure most of these settings, you need to enter your password at the prompt. (The default password is `password`.) After entering the desired settings, click the **Apply** button.

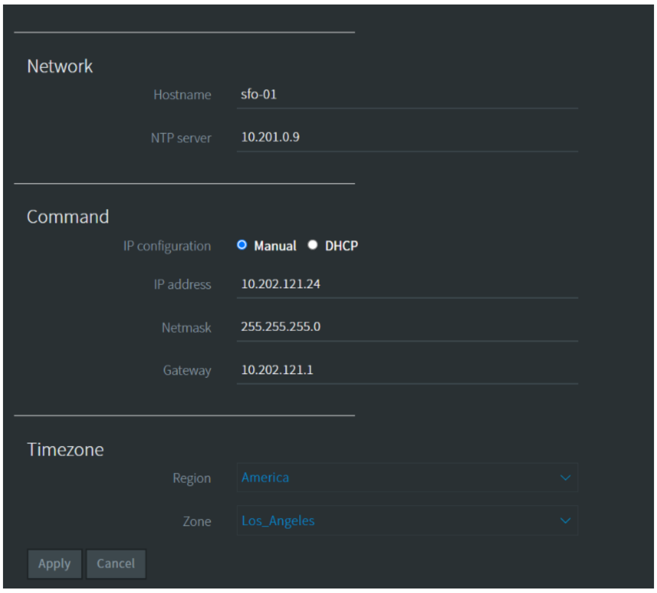


Figure 3-6 Network Screen

3.2.4 Input

To configure the input parameters, click **Input** in the navigation menu. The **Input** screen appears. In this screen, you can select a **digital** or **analog** input type, and configure the **Dolby Atmos Connect** parameters. Starting in system 3.1.0.x the amplifier can now switch between **digital** and **analog** modes and retain the unique settings of each profile. This can be useful in many scenarios.

This section refers to the **digital** input. For analog, proceed to section 3.2.5.

Selecting **digital** tells the amplifier to receive audio via Dolby Atmos Connect Ports (AES67). Selecting **analog** tells the amplifier to receive audio via the 8-channel CAT1416 analog input module (DMA16302 or DMA24302 models only).

This screen defaults to the **Dolby Atmos Connect** > **Manual Mode** input settings which is the default setting for communicating with Dolby Cinema Processors.

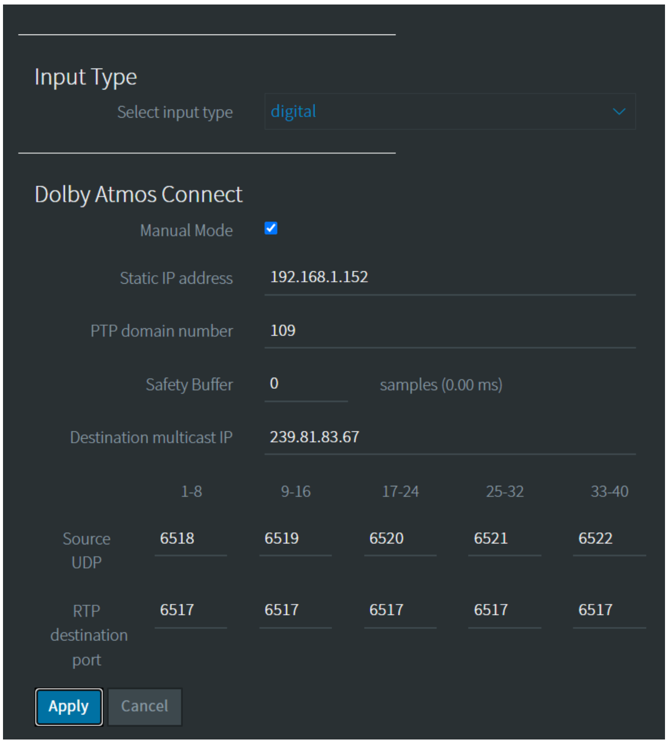


Figure 3-7 Input Screen (digital/manual mode)

The PTP domain number, Destination multicast IP, Source UDP, and RTP destination ports must match with the Dolby Cinema Audio Processor to receive streams.

Alternatively, you can uncheck **Manual Mode** to display the **Available Sessions** mode (also known as discovery mode), where the DMA attempts to discover available AES67 streams from 3rd party devices like the Yamaha Tio or Q-Sys core.

Each **Input Stream** indicates an available 8-channel stream, as shown in the following figure. Use the dropdowns to select from any input stream that is detected by the Dolby Multichannel Amplifier (for example, a Q-Sys core or Yamaha Tio). For details on the Yamaha configuration, see the separate *Dolby CP750 7.1 audio for the Dolby Multichannel Amplifier Installation Manual*.

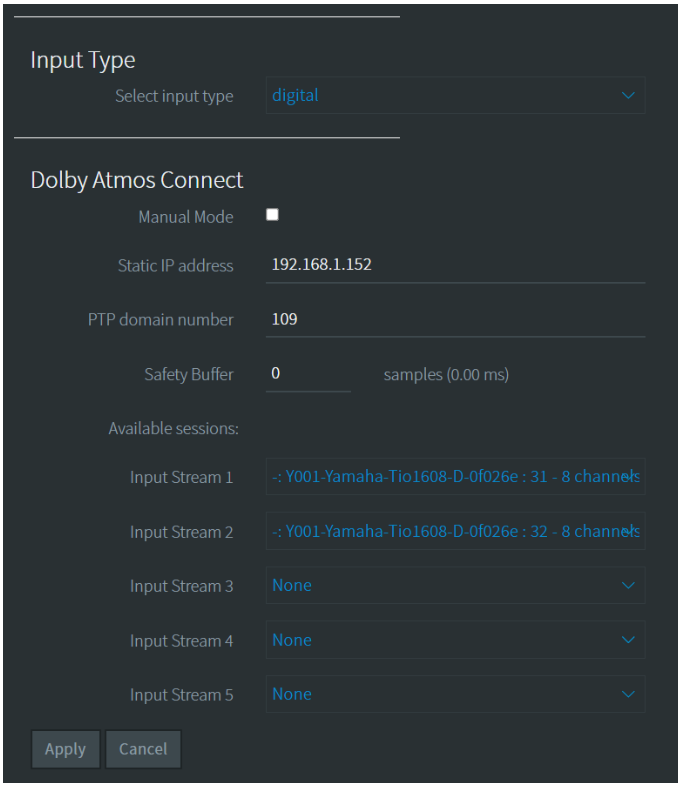


Figure 3-8 Input Screen (digital/discovery mode)

After selecting an available input stream, enter a static IP address. The default PTP domain number is **109**, which you should use for Dolby products. For the Yamaha Tio1608-D input stream configuration use **0**. For help with Q-Sys, contact Dolby Cinema Support. Click **Apply** or **Cancel** to undo.

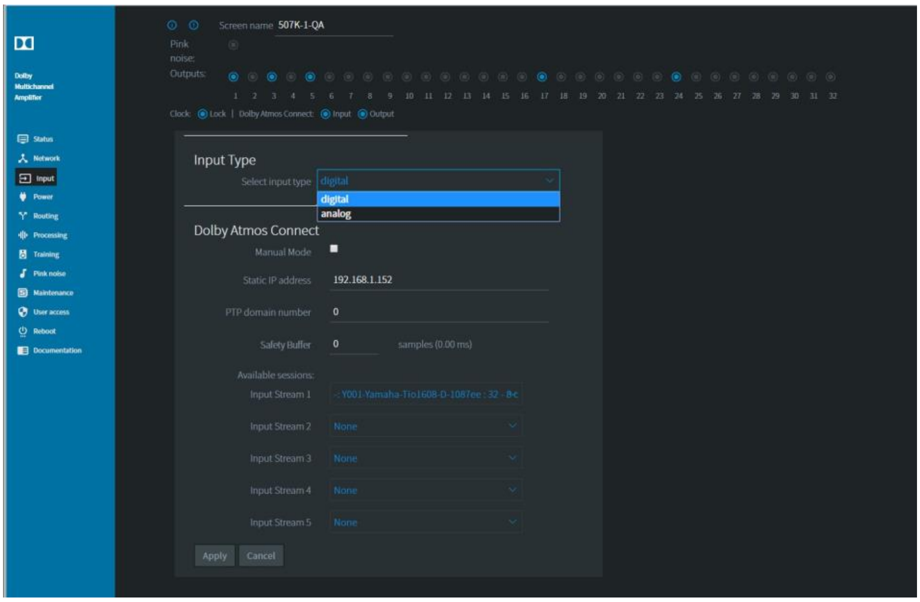
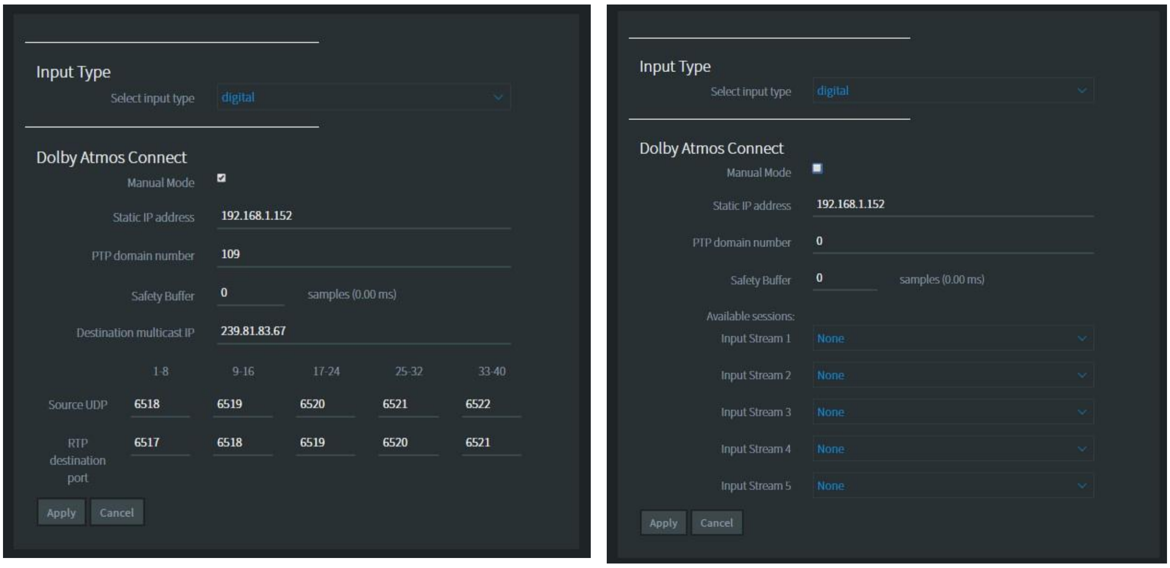


Figure 3-9 Available Sessions Yamaha Tio1608-D Input Stream 1

Dolby Atmos Connect Parameters

The following figure shows the **Dolby Atmos Connect** parameters in digital input manual mode (**Manual Mode** checked) and discovery mode (**Manual Mode** unchecked). With **Manual Mode** unchecked, the amplifier should discover available streams within the current PTP domain. Checking **Manual Mode** will allow you to configure the amplifier to communicate with other digital audio devices using AES67/Dolby Atmos Connect. After entering your settings, click **Apply** or **Cancel** to undo.



Manual mode checked

Manual mode unchecked (discovery mode)

Figure 3-10 Dolby Atmos Connect

Following is a description of the Dolby Atmos Connect **Manual Mode** parameters that you can edit when **Manual Mode** is checked. To change any of these settings, enter the required parameter in the respective field (or click in the corresponding box), and click the **Apply** button. Before you press **Apply** you can undo your changes by clicking the **Cancel** button.

Static IP Address

In this field, you can change the default Dolby Atmos Connect static IP address. In most cases, the cinema audio processor IP address is 192.168.x.151 (where x = auditorium number). The first downstream device IP address should be 192.168.x.152, and the next device should be 192.168.x.153. All IP addresses on this network must be unique.

PTP Domain Number

Precision Time Protocol (PTP) maintains the audio synchronization between the playback device and any connected devices within an auditorium, such as a Dolby Cinema Processor, Dolby Multichannel Amplifier, or DAC3202. The default setting is **109**. If only one auditorium within a facility has a Dolby Multichannel Amplifier installed, we recommend that you retain this default value. If more than one auditorium is interconnected on the same network, you must enter unique PTP values for each auditorium. The PTP domain number in this field must match the PTP domain number in the CP950/A, CP850, IMS3000, or another playback device.

In some situations, third-party devices may use as setting of **0** which may be used for auto-discovery of streams.

The Dolby Multichannel Amplifier does not allow you to set the PTP clock priority. It is always 255 (or the lowest priority in the chain). It should never be chosen as the source of the clock for the network. It is strictly a downstream device that is looking to another clock source.

Safety Buffer

When necessary, you can use the **Safety Buffer** parameter to increase the Dolby Atmos Connect input buffer. Typically, this parameter should remain at 0. However, if you experience network issues that affect audio (brief audio dropouts), you can change this parameter. Running the minimum value to resolve the issue is best. Click in the **Safety Buffer** field, enter a value for up to 50 samples (1.04 milliseconds), and then click **Apply**. If you need more than 50 samples, there is most likely another network issue that should be resolved.

Destination Multicast IP

The destination multicast IP address sets the broadcasting IP for the audio in the auditorium. The playback device (Dolby Cinema Audio Processor) multicasts the audio. All downstream units in the auditorium will have the same Destination Multicast IP address entered in this field will simultaneously process the audio. The destination multicast IP address in this field must match the Destination Multicast IP address in the Dolby Cinema Audio Processor. If other auditoriums have Dolby Atmos Connect networks, they should be kept separate by using different Destination Multicast IP addresses (and different PTP domains).

Source UDP and RTP Destination Ports

The cinema audio processor (CP850, CP950/A, or IMS3000) transmits audio in streams of eight channels over AES67. The system sends audio packets with two important values, RTP source and RTP destination user datagram protocol (UDP) ports. Downstream devices with matching values accept these audio packets and ignore packets where one or both values do not match. You can view the respective input fields by checking **Manual Mode**, as shown in the following figure.

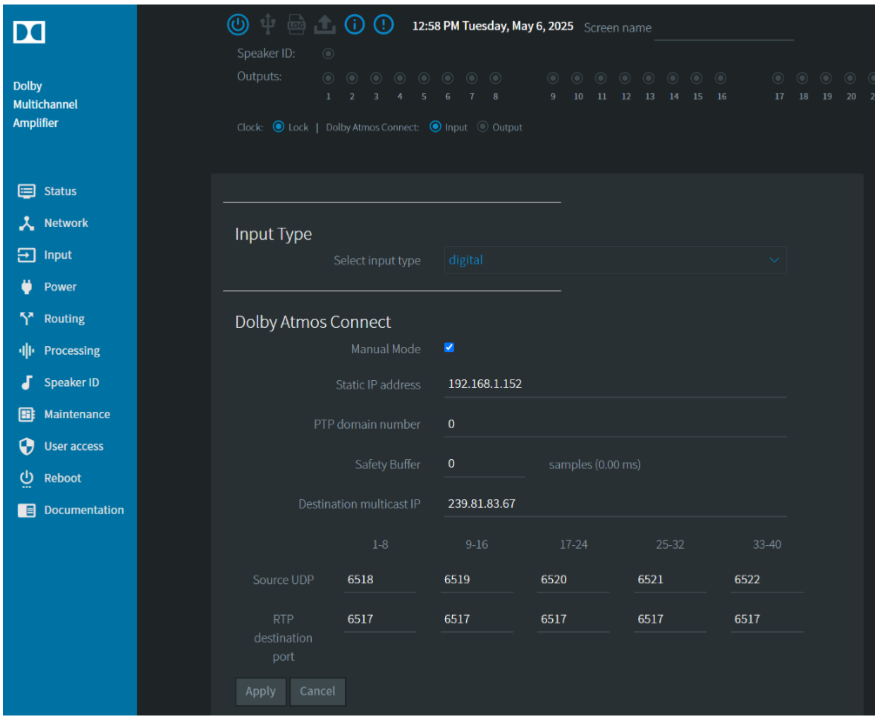


Figure 3-11 Manual Mode Checked

You can enter the desired port numbers for source and destination that is used in your setup.

The default settings are:

Channels	Source UDP port	RTP Destination port
Channels 1–8	6518	6517
Channels 9–16	6519	6517
Channels 17–24	6520	6517
Channels 25–32	6521	6517
Channels 33–40	6522	6517
Channels 41–48	6523	6517
Channels 49–56	6524	6517
Channels 57–64	6525	6517

3.2.5 Using the analog inputs on the Dolby Multichannel Amplifier

To select an analog **Input Type**, click in the **Select input type** field and select **analog**.

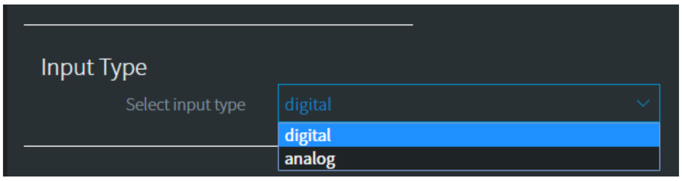


Figure 3-12 Select Input Type

Note: When changing **input type**, the Power (bridging), Routing, channel configuration and Processing settings for each type will be saved and reapplied when switching back. The saving of profiles is new in system 3.1.0.xx and is not possible in previous versions of amplifier software. If you are planning to downgrade the software to a version that does not support managing the different profiles, please back up the DMA in each profile before downgrading.

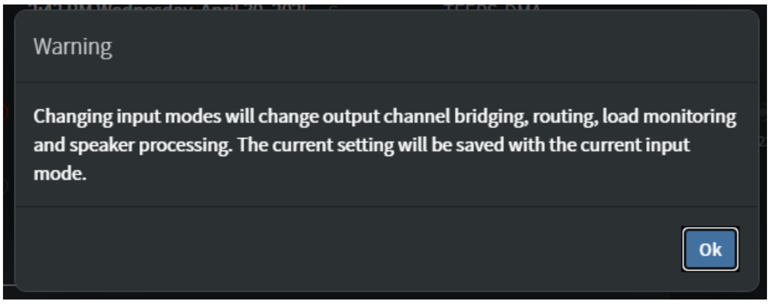


Figure 3-13 Input Type Warning Message

To connect to the amplifier's analog inputs from Dolby audio processors with analog outputs (CP750 with the THX pinout), your Dolby Multichannel Amplifier must have a CAT1416 analog input module installed. The CAT1416 receives up to eight channels of balanced analog audio.

3.2.6 Power

To configure the power parameters, click **Power** in the navigation bar. The Dolby Multichannel Amplifier determines the input voltage (line voltage) automatically but is not aware of the circuit breaker rating (15, 16, or 20 amps).

In this screen, you can set the circuit breaker parameters, enable/disable bridging for the desired channel pairs, set how the unit responds to a loss of power, define the standby parameters, and configure channels for open load monitoring (detecting damaged/missing wiring or speaker elements).

After entering your settings, click **Apply** (or **Cancel** to undo).

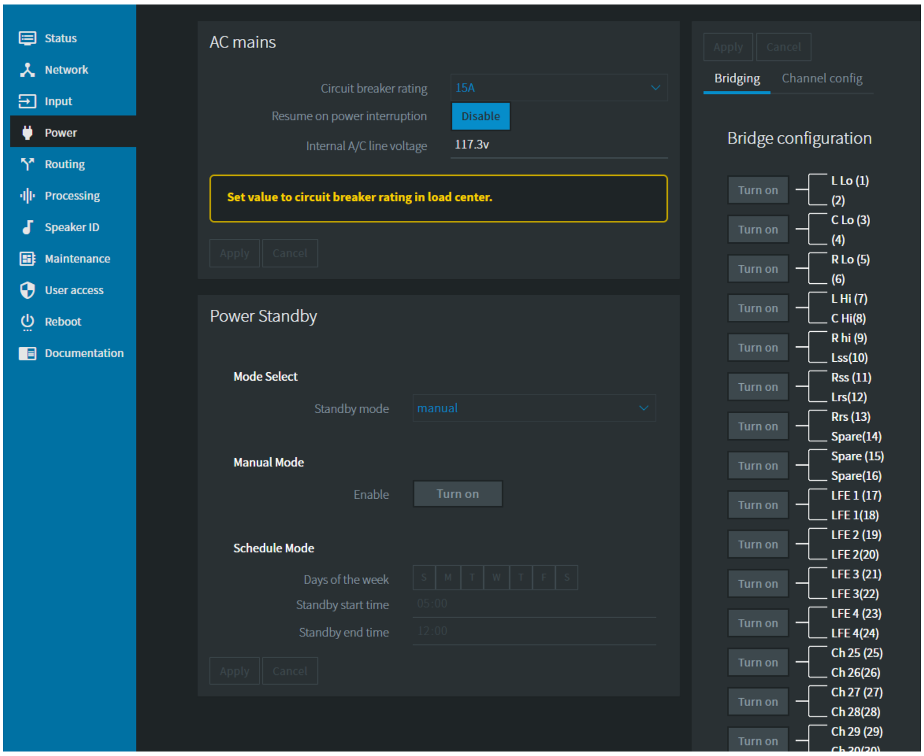


Figure 3-14 Power Screen

Bridging

This option allows you to configure bridging for Dolby Multichannel Amplifier outputs by hovering the mouse over the **Turn on** button next to a channel pair. The corresponding button will then illuminate in blue. You click on this button to turn on bridging and the button changes to **Turn off**, in which case, you click on it to turn bridging off.

After entering your settings, click **Apply** (or **Cancel** to undo).

Note: You must configure any bridged channels before setting up your Routing assignments.

Note: For systems with version 1.0.x.x software, bridging channels results in a loss of the even numbered output for use on other speakers. For example, when you bridge channels 1 and 2, you cannot use channel 2. Upgrading to version 2.x.x software or higher provides improved routing options that allow you to use the even numbered channels.

Resume On Power Interruption

When you check this option, the system enables the Dolby Multichannel Amplifier to automatically turn itself on if the AC mains input power is interrupted. Once the AC mains input is reapplied, the unit automatically turns on and there is no need to press the front-panel power button.

Power Standby Mode

This feature allows you to put the amplifier in a power-saving standby mode, either manually or by a schedule, resulting in an approximate 50% savings in power usage.

By default, the system is in “Manual Mode”. Requiring user intervention to put the unit into Standby mode. This can be done by clicking the power button at the top of the user interface or from the **Power** tab, you can click the Enable button in “Manual Mode” section, then click apply.

The other mode is “Schedule Mode”. For this mode to work, the amplifier must be connected to a valid NTP (Network Time Protocol server) as the amplifier does not have an internal clock.

In either Manual or Schedule mode, you can disable standby mode by:

- Press the yellow blinking power button at the top of the web UI. Then select “apply” to approve the status change.
- On the **Power** tab in the web UI, find the “Manual Mode” section and select “Turn off”. Then click “apply” to approve the status change.
- A long-press (2-seconds) of the front panel power button (which will be flashing orange)

Note: If you are running midnight or special screenings after hours, be sure to switch the amplifier to manual mode so that it does not enter standby mode during the show.

While in Standby Mode, an orange warning indicator will flash on the web UI and the front panel power button will flash orange as well. No setting changes will be allowed in Standby Mode.

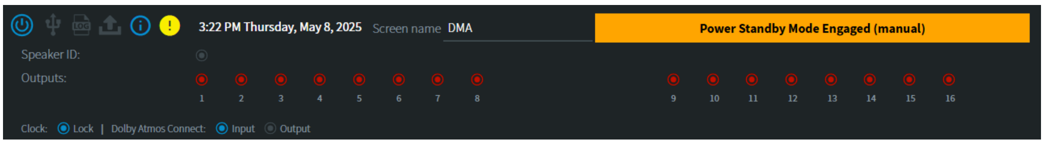


Figure 3-15 Power Screen

When attempting to turn on standby mode using the UI button (power) an "are you sure" dialog will be displayed.

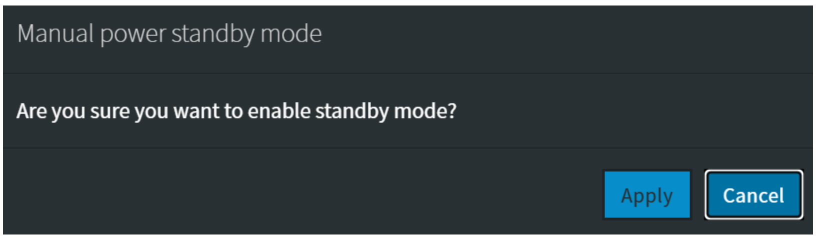


Figure 3-16 Standby mode warning dialog

Schedule Mode

To schedule the amplifier to enter Standby Mode at a predetermined time (ex: when the auditorium is not in use), select Schedule Mode from the dropdown menu.

Time selection is represented using a 24-hour or 12-hour clock (depending on your computer settings) and will follow the time zone settings entered on the **Network** tab. When properly configured and running, the current date and time will be displayed in the header of the WebUI. With no valid NTP connection, schedule mode controls will be disabled, and a warning will be displayed in the Power Standby section.

Select the days of the week you would like to enable standby, as well as the **start time** and **end time**. All selected days will share the same time window. An example is provided below where the amplifier enters standby mode at midnight (00:00) and powers up at 7:00 am.

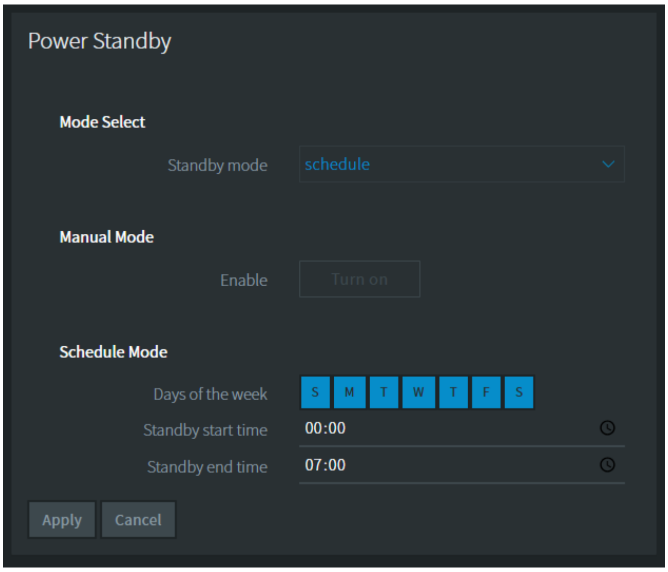


Figure 3-17 Power Standby

Schedule mode will only engage standby if the NTP server is configured and the amplifier is receiving a valid date and time.

Note: If the "end time" is earlier in the day than the "start time" the amplifier will exit standby mode on the following day.

If the amplifier is in schedule-standby mode, the user interface will not allow the unit to manually enter standby mode. An information dialog letting the user know they must navigate to the power page and switch to manual-standby mode first.

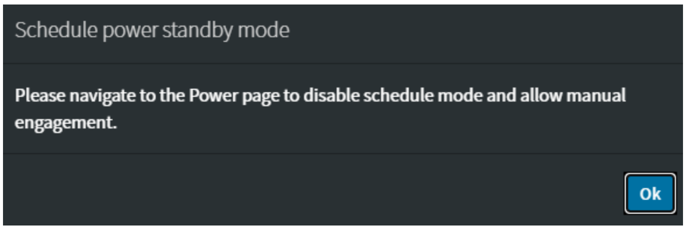


Figure 3-18 Power Standby

SNMP for Standby Mode

Standby mode allows for several new functions via SNMP.

GET operations:

- **StandbyEngaged**: (what is the current system state where 1=Standby and 2=Normal Operation)
- **StandbyMode**: (manual=0 or schedule=1)
- **StandbyScheduleModeTable**: schedule mode state for every day of the week (where a value of 1=schedule engaged for that day or 2=schedule not engaged for that day),
    - Schedule mode start and end times are presented as Hour:Minute strings

SET operations:

- **StandbyMode**: (manual=0 or schedule=1)
- **ManualStandbyEnabled**: (1=on or 2=off)
    - Note this operation will only work if the unit is in manual mode. It does not work when the system is set to schedule mode.
- **StandbyScheduleModeTable**: schedule mode state for every day of the week
    - Schedule mode start and end times.
        - Using a MIB browsing tool may be useful in setting the values for the days (1 or 2) and the Schedule mode start/end times are entered as Hour:Minute strings.
        - When setting start and end times via SNMP, the user must enter 24-hour time format.

Channel Config

In this section of the Power page, you can enable or disable **Load Monitoring** for each speaker that is attached to the amplifier and set the thresholds for proper speaker impedance. This will help the amplifier identify when there might be a problem with the connected speaker. Usually this is either an open or short circuit condition in either the wiring or the speaker driver.

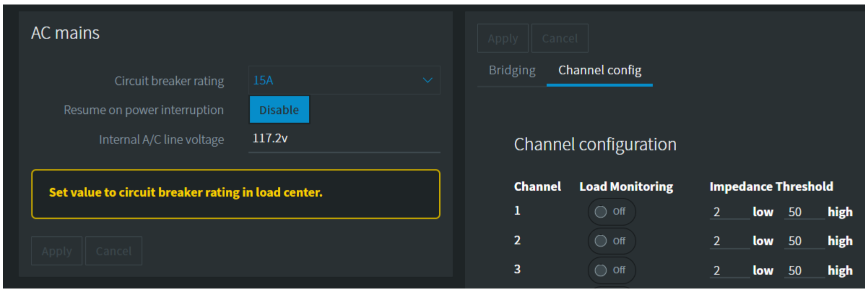


Figure 3-19 Channel config

For each channel, decide whether load monitoring will be used, and then set the low and high impedance thresholds for that speaker. Typically, 2 Ohms is a good low boundary to establish the difference between a short circuit (no impedance) and a low frequency driver (low impedance). The high boundary should be set to something just above the impedance of the speaker's rated impedance. Check the speaker's documentation for confirmation. In some situations, it may be advantageous to input a value several ohms above the speaker's rating to prevent false positive readings that can occur in normal operation.

Note: The status for each channel configured to use Load Monitoring will begin as "grey" or unknown until sufficient audio energy has been provided for the system to start monitoring. This could take several minutes after audio playback has begun.

3.2.7 Routing

To configure the routing parameters, click **Routing** in the navigation bar. The amplifier can receive up to five input streams of eight channels. Input streams and the respective UDP ports appear on the left side of the **Routing** screen, and outputs are listed on top. The UDP ports correspond to the UDP settings in the **Network** > **Dolby Atmos Connect** screen. The number of outputs corresponds to the number of amplifier cards installed. The amplifier is shipped with no routing assignments, and no audio is transmitted until you configure routing.

At the top of the page, you can switch between **Route assignment** and **Wiring config** which shows you a rendering of the back panel of the DMA with the channels labeled.

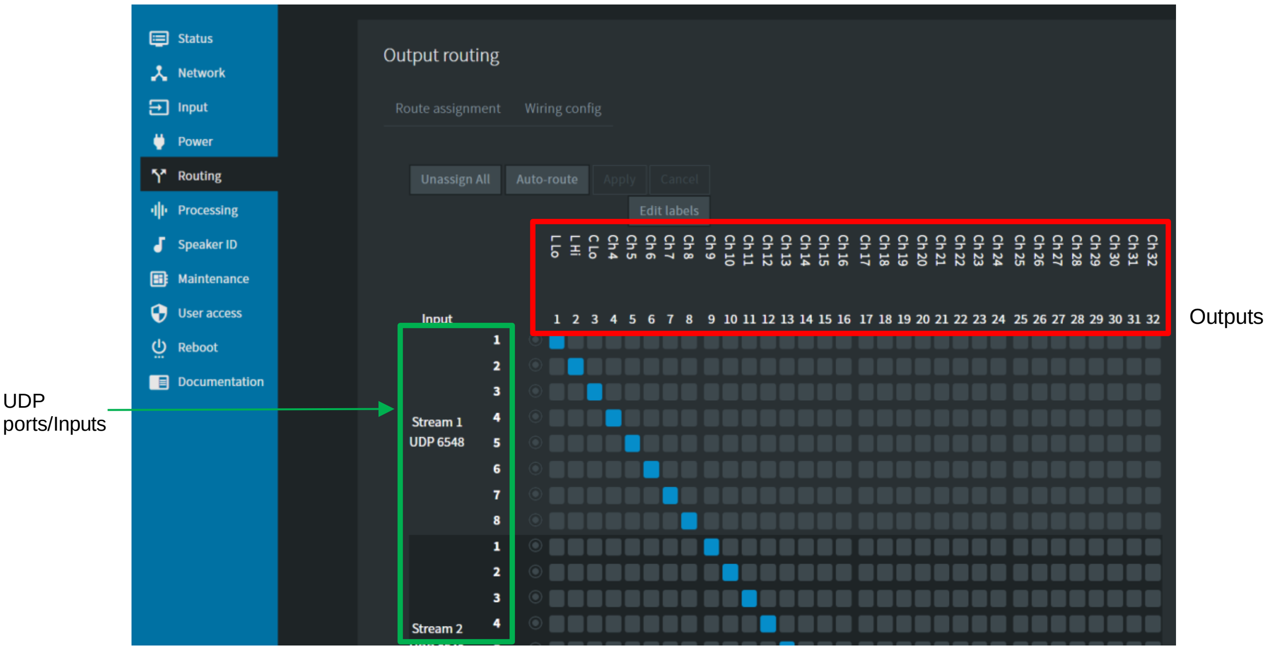
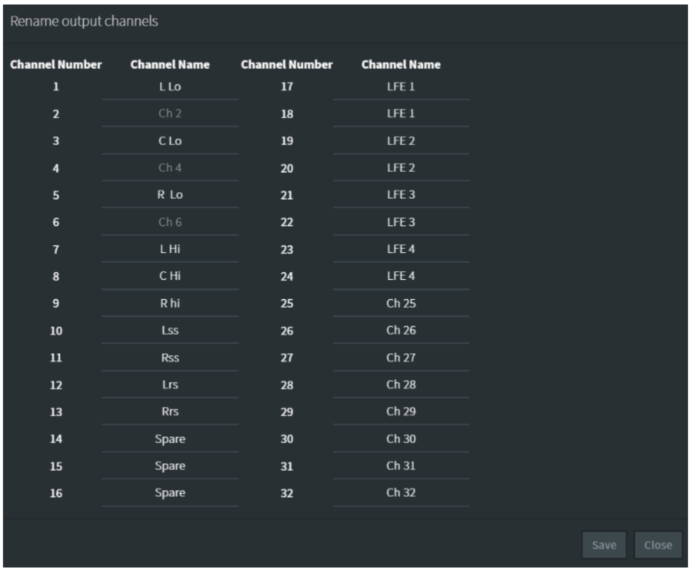


Figure 3-20 Routing Screen (Route Assignment view)

Figure 3-21 Routing Screen (Wiring config view)

You can use **Edit labels** to change the designation of the labels in the **Wiring config** view. The values selected here will be represented on the Status, Power, Processing and Speaker ID pages


Rename output channels

Channel Number	Channel Name	Channel Number	Channel Name
1	L Lo	17	LFE 1
2	Ch 2	18	LFE 1
3	C Lo	19	LFE 2
4	Ch 4	20	LFE 2
5	R Lo	21	LFE 3
6	Ch 6	22	LFE 3
7	L Hi	23	LFE 4
8	C Hi	24	LFE 4
9	R hi	25	Ch 25
10	Lss	26	Ch 26
11	Rss	27	Ch 27
12	Lrs	28	Ch 28
13	Rrs	29	Ch 29
14	Spare	30	Ch 30
15	Spare	31	Ch 31
16	Spare	32	Ch 32

Save Close

Figure 3-22 Routing Screen (Wiring config view)

To configure routing, click in an output box in the row next to the desired input stream; the box is then highlighted in blue to show the corresponding routing output. If you click again in a highlighted box, that assignment is removed. You can assign individual inputs to one or more outputs. However, you cannot assign multiple inputs to an output.

Note: To configure bridged channels, you must select Power in the navigation bar and set up bridging before configuring your routing assignments.

Bridged channels are represented as one output, but input channels are not skipped in the automatic routing map. You can route an input to the first channel of a bridged pair but cannot route an input to the second channel of a bridged pair.

In the next figure, channels 5/6 and channels 7/8 are bridged.

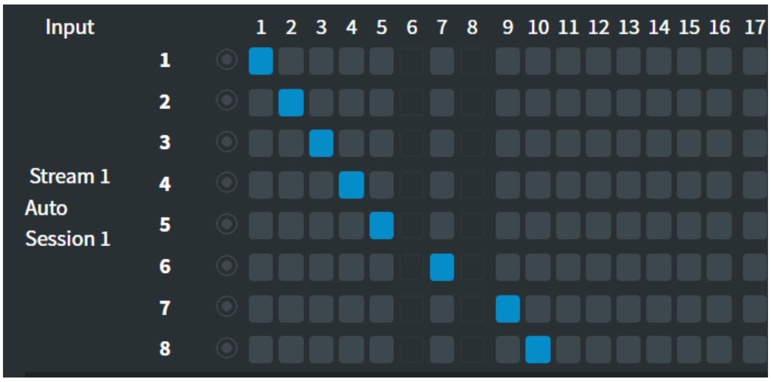


Figure 3-23 Routing with bridged channels

If you click the **Auto-route** button, the system looks for every available input channel and assigns consecutive outputs to consecutive inputs (input 1 to output 1, input 2 to output 2, and so on). When the system sees a bridged channel, a gap will appear (as seen above).

If you click **Unassign all**, the system removes the routing assignments.

Click **Apply** to save your routing assignments or **Cancel** to discard your changes.

CAT1416 Analog Input Routing

When the Dolby Multichannel Amplifier CAT1416 analog input is used (DMA16302 or DMA24302 models only), the **Routing** screen changes to show the available channels used in a standard Dolby Surround 7.1 auditorium.

Signal presence is also represented by lighting up the circle indicators in blue.

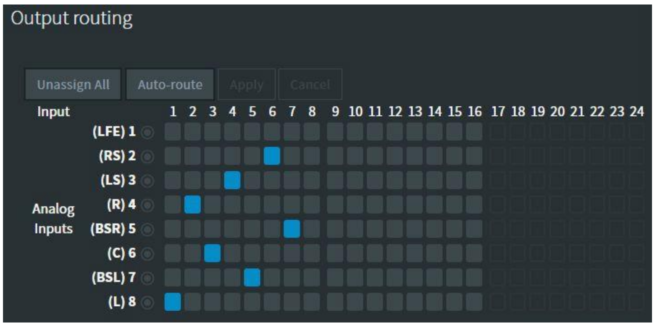


Figure 3-24 Analog Input Routing Screen

The channel labels are always displayed for a Dolby Cinema Processor CP750 that is connected to the optional Dolby DB-25 cable (DMA-ACC-ANA-CBL). The use of other cinema processors or cables with different wiring may not match the channel labels displayed in this screen. Pinout information for the CAT1416 analog input card is provided in Appendix C.

3.2.8 Processing

To configure the speaker processing parameters, click **Processing** in the navigation bar. In this screen, you can configure basic speaker settings for gain, delay, polarity, crossover filters, and mute channels. If you configure a set of parameters and want to copy those parameters to another output, you can do this with **Duplicate Filters.**

Note: When you apply any setting changes, audio is muted briefly.

Speaker Processing

To configure the speaker settings, click in the **Channel** field, select a channel from the **Channel** drop-down menu (or use the left/right arrows above the graph), click **Enable Speaker Processing** in the **Basic Settings** pane to activate these settings, and then click **Apply.** The button changes to **Disable Speaker Processing**, where you can click to turn off speaker processing.

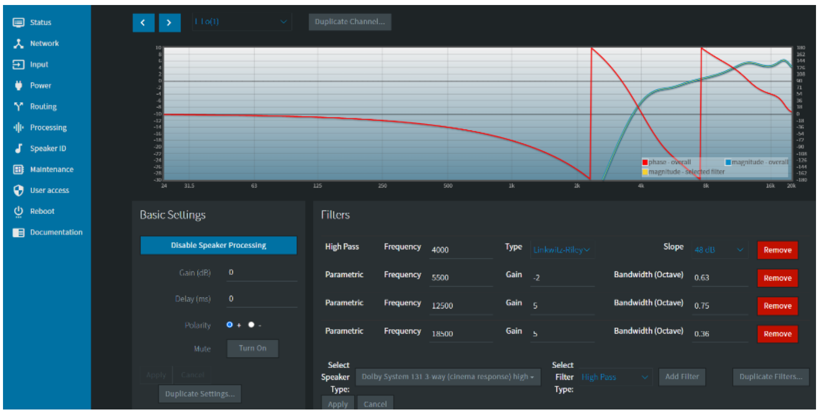


Figure 3-25 Adjust Audio Delay

Following is a description of the speaker processing parameters beginning with **Basic Settings**.

Gain (dB)

You can use the gain parameter to adjust the output gain in 0.1 increments for each output channel. Click in the **Gain (dB)** field and enter a value (or use the up/down arrows) to specify the desired gain within the range of +11 to -100 dB and click **Apply**. The default gain value is 0 dB for current units. Units upgraded from earlier software versions display a default gain value of 10.88 dB. We recommend keeping the gain at or close to 0 dB whenever possible. If gain boost is needed to reach the required level in the auditorium, this should be done temporarily to complete the EQ. Refer to Appendix E for the steps that describe our Gain recommendation.

Delay (ms)

To adjust audio delay, click in the **Delay (ms)** field (or use the up/down arrows) to enter a value to specify the global audio delay for up to 2.6 ms, and then click **Apply**.

Polarity

To change the speaker polarity, click the **+** or **-** buttons, and click **Apply**.

Mute

To mute or unmute a channel, click the **Mute** button to toggle this setting on and off, then click **Apply**.

Duplicate Settings

After configuring the basic settings for Speaker Processing, you can use **Duplicate Settings** to clone the same settings to another speaker.

After **Basic Settings** there is another section called **Filters.**

Filters

To specify a filter type, click in the **Select Filter Type** field, and select the desired filter in the drop-down menu.

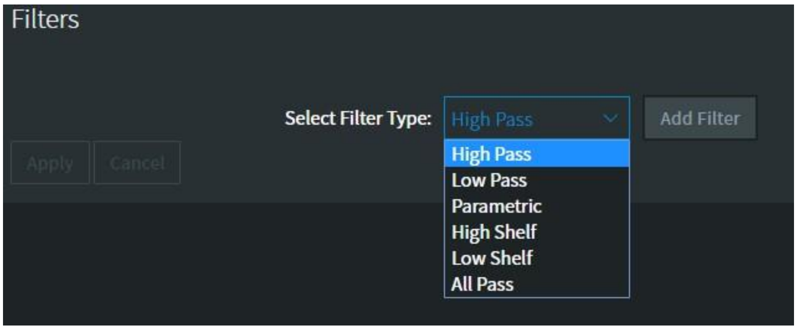


Figure 3-26 Specify Filter Type

Note: **Parametric** and **Shelf** equalization filters include gain and bandwidth (Q), while the **All Pass** filter includes order.

To add a filter, click the **Add Filter** button (which changes to a **Remove** button), enter the desired settings for **Frequency**, **Type**, and **Slope**, then click the **Apply** button.

To delete a filter, click the **Remove** button. To duplicate a filter, click the **Duplicate Filters** button.

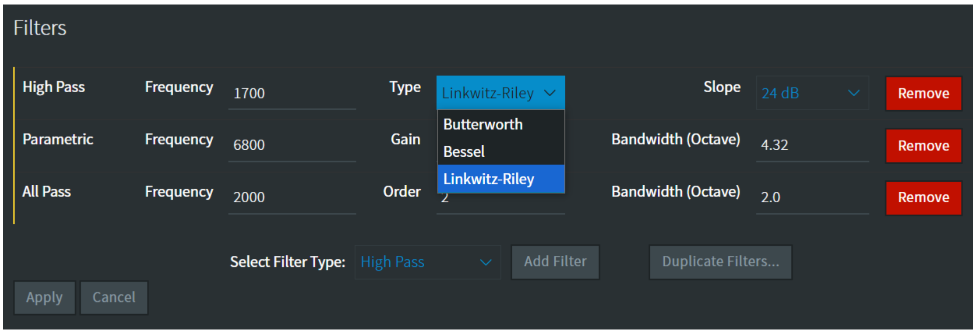


Figure 3-27 Add Filter

3.2.9 Speaker ID

To audibly identify a speaker connection, click **Speaker ID** in the navigation bar. This will play filtered noise (lower high frequency range to protect tweeters) to help identify that a speaker is connected to the correct terminals on the DMA. Note this signal should not be used for EQ'ing the auditorium.

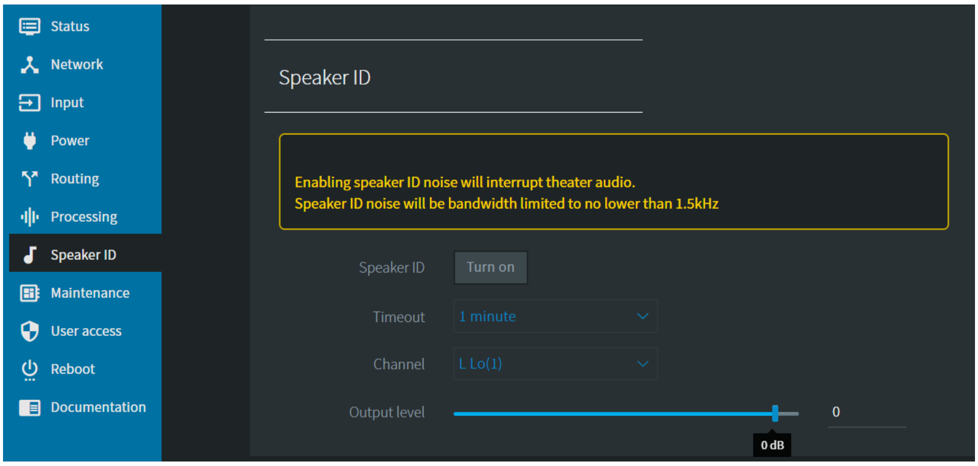


Figure 3-28 Speaker ID Noise Screen

The Dolby Multichannel Amplifier can output a **Speaker ID** Noise (high frequency filtered noise) to an auditorium, which enables you to verify speaker connections and basic operations without a playback device. To turn on **Speaker ID**, select the desired channel and **Speaker ID** time-out interval (1-15 minutes), and use the slider to adjust the **Speaker ID** output level. Then click the **Turn on** button to generate **Speaker ID** Noise. The **Speaker ID** indicator at the top of the screen illuminates in blue and starts the desired countdown.

Note: The Dolby Multichannel Amplifier **Speaker ID** stimulus is noise with a High Pass 24 dB Butterworth filter applied at 1.2 kHz. You should use care to ensure that the Dolby Multichannel Amplifier output level does not cause over excursion to the attached loudspeakers.

To disable **Speaker ID**, click the red **Turn off now** button at the top of the screen or the red **Turn off** button. When set to 0 dB, the Dolby Multichannel Amplifier outputs **Speaker ID Noise** with a typical RMS output voltage of approximately 1.3 Vrms. Increasing the maximum output to +10 dB will result in 2 Vrms.

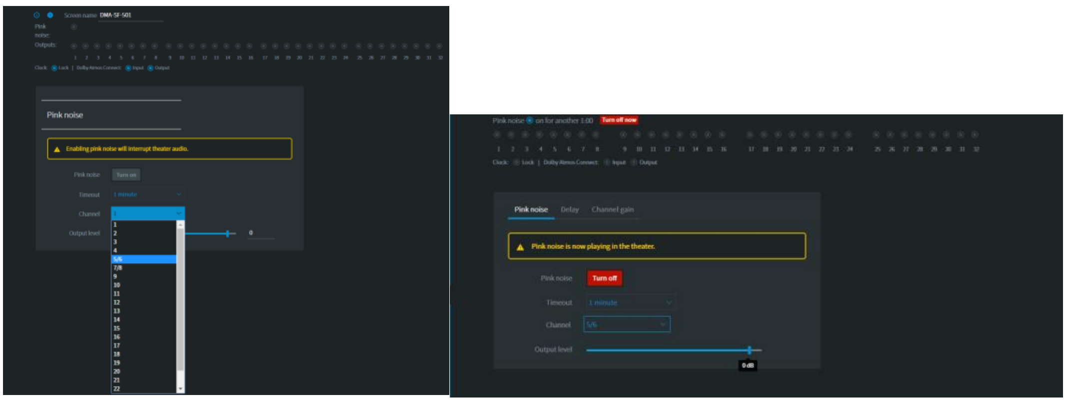


Figure 3-29 Turn On Speaker ID

3.2.10 Maintenance

To perform maintenance tasks, click **Maintenance** in the navigation bar.

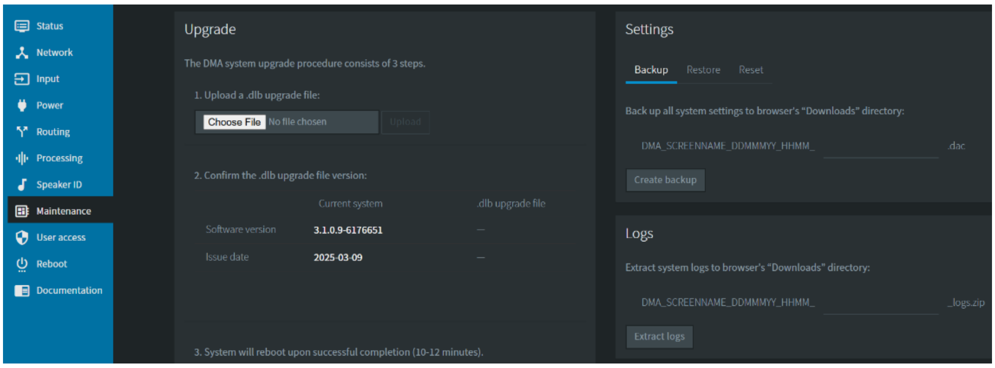


Figure 3-30 Maintenance Screen

Following is a description of all the maintenance tasks.

Upgrade

To upgrade your Dolby Multichannel Amplifier system software:

1. Obtain the .dlb upgrade file from Dolby Laboratories at https://customer.dolby.com/cinema/
2. Copy the upgrade file to your local disk.
3. Click **Choose File** to browse to the upgrade file, and then click the **Upload** button. The uploading icon  flashes at the top of the screen.
4. Confirm the displayed upgrade file version, and then click the **Install Upgrade** button. If you need to cancel, click the **Cancel** button.
5. Enter your password at the prompt. (The default password is `password`.)

The system initiates the upgrade process, displays the progress, and reboots the Dolby Multichannel Amplifier when the upgrade is completed.

Note: After upgrading to 3.1.x for the first time, there are some functions that will not work until a repeat upgrade is completed to the same version. This is a one-time operation needed when crossing the threshold from 3.0.x to 3.1.x. The missing functions are:

- Documentation – When you click on any of the documents, no file will be downloaded.
- The ability to perform any of the following Alternative USB maintenance functions when booting up the amplifier. The only time that Alternative USB functions (upgrade, restore, backup, and logs) will work is when the system is fully booted.

Settings

To back up your system settings:

1. Click the **Backup** button and enter a name for your .dac restore file.
2. Click the **Create backup** button to save the .dac restore file in your browser downloads directory on your PC.

To restore your system settings:

1. Click the **Restore** button, then click **Choose File** to select a .dac restore file. You can select from **All settings** or four individual restore options:
   - **All settings**
     - **Network tab settings only**
     - **AC Mains tab settings only**
     - **Bridge Configuration**, **Channel configuration and Output routing tab settings only**
     - **Speaker Processing Basic settings and Filter tab settings only**).
2. Click the **Upload and restore** button to run the restoration process.

If you need to cancel, click the **Cancel** button.

To reset your system to the factory defaults (except for network tab settings).

1. Click the **Reset** button.
2. Click the second **Reset** button at the lower-left side of the settings pane to run the reset operation.

If you need to cancel, click the **Cancel** button.

Logs

To download system logs, enter a name for your logs .zip file, and then click the **Extract logs** button to save a .zip file containing the logs in your browser downloads directory on your PC. The log icon flashes at the top of the screen while the logs are being extracted.

Alternative Maintenance Procedures

A dedicated Dolby Multichannel Amplifier maintenance application, separate from the web client user interface, is provided for cases where there may be system issues. In such a case, you can use a USB drive to upgrade the system, backup and restore settings, and extract logs. A flashing USB icon appears at the top of the Web UI when a USB drive is connected.

Note: After upgrading to 3.1.x for the first time, there are some functions that will not work until a repeat upgrade is completed to the same version. This is a one-time operation needed when crossing the threshold from 3.0.x to 3.1.x. The missing functions are:

- Documentation – When you click on any of the documents, no file will be downloaded.
- The ability to perform any of the following Alternative USB maintenance functions when booting up the amplifier. The only time that Alternative USB functions (upgrade, restore, backup, and logs) will work is when the system is fully booted.

Note: If you are using a cable to connect a USB drive, be sure to use a shielded USB cable.

Alternative USB Upgrade Procedure

To perform this operation:

1. Obtain the .dlb upgrade file from Dolby Laboratories at https://customer.dolby.com/cinema/, and copy it to the root (top) level of a USB drive. Supported file systems are vFAT, FAT32, and FAT16.

We recommend that your USB drive does not include any other files or folders.

2. Insert the USB drive into the Dolby Multichannel Amplifier USB port, and then reboot the unit by clicking **Reboot** in the navigation bar.

The system mounts the drive and checks for the presence of the upgrade .dlb file. If a .dlb file is present, the system compares the version number of the file to the currently installed version. If the currently installed version is older, the system starts the upgrade. After the upgrade is complete, the system automatically reboots.

If you insert a USB drive after the boot process begins, no upgrade is performed. The USB-based upgrade is performed only when the drive is inserted prior to booting up.

If the upgrade fails, the system performs the backup/restore/logs extract process, as described in the next section, and then unmounts the drive.

Alternative USB Backup, Restore, and Extract Logs Procedure

When you insert a USB drive that does not contain an upgrade file, the system will automatically:

- Download a backup of the system settings to a **dma_backup.dac** file on the USB drive.
- Extract a copy of the system logs to the USB drive.
- Restore system settings to factory defaults if you save a text file with the name **dma_restore_defaults**, remove the .txt extension, and include the file on the USB drive.
- Restore system settings from a **dma_backup.dac** file if you change the name of that file to **dma_restore.dac** and include it on the USB drive.

To perform the above tasks, insert a USB drive into the Dolby Multichannel Amplifier USB port. The system mounts the drive and automatically performs the following operations:

- Backs up the current settings on the USB drive. The settings file is named **dma_backup<N>.dac**, where N is a sequence number. The sequence number is determined as the system scans the drive to detect any previously created backup files. If none are found, the system creates dma_backup0.dac. If previous backups are found, the system uses the next number in the sequence.
- Extracts logs to a .zip file using the same sequence scheme as for backing up settings: dma_logs0.zip, dma_logs1.zip, and so on.
- Checks the USB drive for a file named **dma_restore_defaults**. If the system finds this file, the system settings are restored to the factory defaults.
- If no **dma_restore_defaults** text file was found, the system checks the USB drive for a **dma_restore.dac** file. If such a file is found, the system restores your backup settings from the .dac file.

After the system restores any settings, the main Dolby Multichannel Amplifier application rereads the settings and applies them. In addition, the system saves a file with the word “complete” in the file name on the USB drive to confirm the restoration. The system then deletes the other restore files from the USB drive to avoid repeating the restore operation.

Note: The USB port LED on the Dolby Multichannel Amplifier flashes during USB access.

Do not remove the USB drive while the LED is flashing.

At the end of this procedure, the system unmounts the USB drive and the USB port LED will become solid blue, indicating that the reading/writing of the USB device is complete.

3.2.11 User Access

To configure user access, click **User Access** in the navigation bar to display the password control screen. To change the password on the amplifier, enter the new password in the corresponding field, enter it again in the **Confirm password** field, and then click **Change password**.

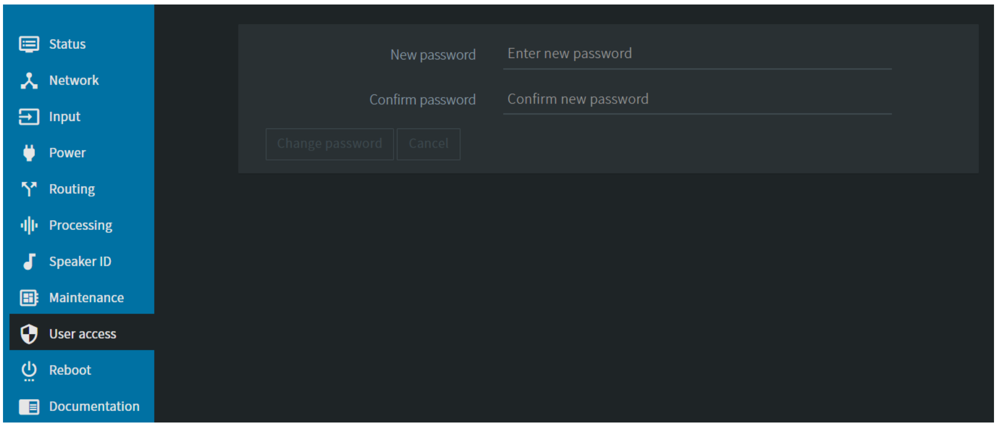


Figure 3-31 User Access Screen

3.2.12 Reboot

To reboot the Dolby Multichannel Amplifier, click **Reboot** in the navigation bar, enter your password at the prompt (or to cancel, click the **Cancel** button) and then click the **Reboot** button.

3.2.13 Documentation

To access Dolby Multichannel Amplifier user documentation, click **Documentation** in the navigation bar.

Note: After upgrading to 3.1.x or later, documentation may not be available. You can restore access to documentation by re-upgrading the amplifier to the latest version.

Specifications

Following are the Dolby Multichannel Amplifier specifications.

A.1 AC Line Current Draw and Thermal Dissipation Specifications

		DMA16301 DMA16302	120 VAC			208 VAC			230 VAC		
Output Level	**Load**	**Output Configuration**	**Line Current (A, rms)**	**Dissipated Power (Watts, rms)**	**BTU/ Hour**	**Line Current (A, rms)**	**Dissipated Power (Watts, rms)**	**BTU/ Hour**	**Line Current (A, rms)**	**Dissipated Power (Watts, rms)**	**BTU/ Hour**
Idle	-	-	1.86	214	730	1.4	205	699	1.3	199	679
1/8 power pink noise	2Ω	Normal	16.2	674	2,300	9.4	659	2,249	8.4	638	2,177
	4Ω	Normal	16.0	639	2,180	9.2	606	2,068	7.9	550	1,877
	4Ω	Bridged	17.0	858	2,928	9.3	754	2,573	8.3	725	2,474
	8Ω	Normal	8.6	404	1,379	4.9	385	1,314	4.4	359	1,225
	8Ω	Bridged	14.9	639	2,180	8.6	591	2,017	7.6	565	1,928
1/4 power pink noise	2Ω	Normal	See Note 5			17.9	1,169	3,989	16.3	1192	4,067
	4Ω	Normal				16.7	889	3,033	14.8	845	2,883
	4Ω	Bridged				17.2	1,191	4,064	15.6	1,213	4,139
	8Ω	Normal	14.7	506	1,727	8.3	461	1,573	7.5	448	1,529
	8Ω	Bridged	See Note 5			15.9	929	3,170	14.2	888	3,030

		DMA24302	120 VAC			208 VAC			230 VAC		
Output Level	**Load**	**Output Configuration**	**Line Current (A, rms)**	**Dissipated Power (Watts, rms)**	**BTU/ Hour**	**Line Current (A, rms)**	**Dissipated Power (Watts, rms)**	**BTU/ Hour**	**Line Current (A, rms)**	**Dissipated Power (Watts, rms)**	**BTU/ Hour**
Idle	-	-	2.2	258	880	1.5	251	856	1.5	254	867
1/8 power pink noise	2Ω	Normal	See Note 5			13.6	956	3,261	12.2	925	3,157
	4Ω	Normal				13.3	879	2,999	11.4	729	2,529
	4Ω	Bridged				13.5	1,093	3,731	12.0	1,051	3,587
	8Ω	Normal	12.4	544	1,856	7.1	558	1,905	6.2	469	1,369
	8Ω	Bridged	See Note 5			12.5	857	2,925	11.0	819	2,796
1/4 power pink noise	2Ω	Normal				See Note 5					
	4Ω	Normal									
	4Ω	Bridged									
	8Ω	Normal				12.0	668	2,281	10.9	650	2,217
	8Ω	Bridged	See Note 5								

DMA32301			120 VAC			208 VAC			230 VAC		
Output Level	**Load**	**Output Configuration**	**Line Current (A, rms)**	**Dissipated Power (Watts, rms)**	**BTU/ Hour**	**Line Current (A, rms)**	**Dissipated Power (Watts, rms)**	**BTU/ Hour**	**Line Current (A, rms)**	**Dissipated Power (Watts, rms)**	**BTU/ Hour**
Idle	-	-	2.7	321	1,095	1.8	320	1,092	1.7	319	1,088
1/8 power pink noise	2Ω	Normal	See Note 5			18.8	2,518	8,592	16.8	2,476	8,448
	4Ω	Normal				18.4	2,412	8,230	15.8	2,300	7,848
	4Ω	Bridged				18.6	2,608	8,899	16.6	2,250	8,701
	8Ω	Normal	17.2	1,408	4,804	9.8	1,370	4,675	8.8	1,318	4,497
	8Ω	Bridged	See Note 5			17.2	2,282	7,787	15.2	2,230	7,609
1/4 power pink noise	2Ω	Normal				See Note 5					
	4Ω	Normal									
	4Ω	Bridged									
	8Ω	Normal				17.2	2,255	7,694	15.6	2,228	7,602
	8Ω	Bridged	See Note 5								

Notes:

1. Pink noise stimulus with 12 dB crest factor, band-limited 20 Hz to 20 kHz.
2. Data based on all driven channels.
3. Fractional output levels are based upon rated channel power for the given load impedance. (For example, for a 1/8 power 8-ohm normal configuration, the net output power is 300 W/8 x 32 channels = 1,200 W.)
4. Specifications are based on laboratory measurements and should be considered typical values, as they do not constitute absolute limits.
5. Pink noise tests for this configuration are limited by duration due to AC mains breaker rating. Amplifier output limiting occurs to reduce current draw.
6. Specifications for the DMA32301 were derived from the DMA16301 measurements.

DMA32300			120 VAC			208 VAC			230 VAC		
Output Level	**Load**	**Output Configuration**	**Line Current (A, rms)**	**Dissipated Power (Watts, rms)**	**BTU/ Hour**	**Line Current (A, rms)**	**Dissipated Power (Watts, rms)**	**BTU/ Hour**	**Line Current (A, rms)**	**Dissipated Power (Watts, rms)**	**BTU/ hour**
Idle	-	-	2.1	227	775	1.5	227	775	1.5	227	775
1/8 power pink noise	4Ω	Normal	15.2	576	1,965	8.8	548	1,870	7.8	540	1,843
	8Ω	Normal	15.1	567	1,935	8.6	525	1,791	7.7	518	1,767
	8Ω	Bridged	15.7	631	2,153	8.8	550	1,877	8.1	606	2,068
1/4 power pink noise	4Ω	Normal	See Note 5			15.9	859	2,931	14.5	816	2,784
	8Ω	Normal				15.1	701	2,392	13.8	646	2,204
	8Ω	Bridged				16.4	950	3,242	14.8	898	3,064

		DMA24300	120 VAC			208 VAC			230 VAC		
Output Level	**Load**	**Output Configuration**	**Line Current (A, rms)**	**Dissipated Power (Watts, rms)**	**BTU/ Hour**	**Line Current (A, rms)**	**Dissipated Power (Watts, rms)**	**BTU/ Hour**	**Line Current (A,rms)**	**Dissipated Power (Watts,rms)**	**BTU/ Hour**
Idle	-	-	1.8	196	689	1.5	196	669	1.4	196	669
1/8 power pink noise	4Ω	Normal	11.4	432	1,474	6.6	411	1,402	5.9	405	1,382
	8Ω	Normal	11.3	425	1,451	6.5	394	1,344	5.8	389	1,326
	8Ω	Bridged	11.8	473	1,615	6.6	413	1,408	6.1	455	1,551
1/4 power pink noise	4Ω	Normal	See Note 5			11.9	644	2,198	10.9	612	2,088
	8Ω	Normal				11.3	526	1,794	10.3	485	1,653
	8Ω	Bridged				12.3	713	2,431	11.1	674	2,298

		DMA16300	120 VAC			208 VAC			230 VAC		
Output Level	**Load**	**Output Configuration**	**Line Current (A, rms)**	**Dissipated Power (Watts, rms)**	**BTU/ Hour**	**Line Current (A, rms)**	**Dissipated Power (Watts, rms)**	**BTU/ Hour**	**Line Current (A,rms)**	**Dissipated Power (Watts, rms)**	**BTU/ Hour**
Idle	-	-	1.5	152	519	1.3	152	519	1.3	152	519
1/8 power pink noise	4Ω	Normal	7.7	291	993	4.4	277	944	4.0	273	930
	8Ω	Normal	7.6	286	977	4.4	265	905	3.9	262	893
	8Ω	Bridged	7.9	319	1,087	4.4	278	948	4.1	306	1,044
1/4 power pink noise	4Ω	Normal	14.3	408	1,393	8.0	434	1,480	7.3	412	1,406
	8Ω	Normal	13.3	322	1,100	7.6	354	1,208	7.0	326	1,113
	8Ω	Bridged	14.3	408	1,393	8.3	480	1,637	7.5	453	1,547

Notes:

1. Pink noise stimulus with 12 dB crest factor, band-limited 20 Hz to 20 kHz.
2. Data based on all driven channels.
3. Full power = 32 channels at 300 Wrms per channel (same net power for bridged mode; 16 channels at 600 Wrms per channel).
4. Specifications are based on laboratory measurements and should be considered typical values, as they do not constitute absolute limits.
5. Pink noise tests for this configuration are limited by duration due to AC mains breaker rating. Amplifier output limiting occurs to reduce current draw.
6. Specifications for the DMA24300 and DMA16300 are mathematically derived from the laboratory measurements made on the DMA3200.

A.2 DMA16301, DMA16302, DMA24302, and DMA32301 Audio Specifications

Parameter	Typical Performance Specification			Measurement Notes
Power output rating	**Unbridged**	**Bridged**		**Dolby power amplifier rating specifications:** 1. Burst: 1 kHz for 20 ms, 10 kHz for 10 ms, two channels driven 2. Short term: 20 Hz, 1 kHz, and 20 kHz at -1 dB for five seconds, two channels driven 3. Long term: 1/8th power pink noise for one hour, two channels driven
	300 watts	1,100 watts	8Ω	
	600 watts	1,100 watts	4Ω	
	600 watts	NA	2Ω	
Power budget rating (total audio power available) Full-range	120 VAC	208 VAC	230 VAC	**For full-range specification:** 1. Burst 50Hz for 200 ms, 1 kHz for 20 ms, 10 kHz for 10 ms 2. Short term 20 Hz, -1 dB for five seconds 3. Total power summed across all channels driven prior to any limiting
	1,980 watts	3,480 watts	3,480 watts	
THD+N (1 kHz)	0.004%–0.009%		8Ω	1 dB below rated power, AES-17 20 kHz LPF, two adjacent channels driven in normal mode
	0.009%–0.02%		4Ω	
THD+N (20 Hz to 20 kHz)	0.05%		8Ω	
	0.10%		4Ω	
Frequency response	20 Hz to 20 kHz, +0.4 dB/–0.2 dB		8Ω	
Intermodulation distortion ratio (SMPTE 4:1)	0.05%			1dB below rated power, SMPTE 4:1 60 Hz and 7 kHz, AES17 20 kHz lowpass filter
Signal-to-noise ratio	109 dB			A-weighted, AES17 20 kHz lowpass filter
Channel separation crosstalk	70 - 90 dB		8Ω	Depending upon channel utilization, measured at 1 kHz
DC offset	<±5 mV			
Output impedance	44 mΩ			
Damping factor	180		8Ω	Measured 20 Hz to 1 kHz

A.3 DMA32300, DMA24300, and DMA16300 Audio Specifications

Parameter	Typical Performance Specification			Measurement Notes
	Unbridged	**Bridged**		
Power output rating	300 watts	600 watts	8Ω	**Dolby power amplifier rating specifications:** 1. Burst: 1 kHz for 20 ms, 10 kHz for 10 ms, half channels driven 2. Short term: 20 Hz, 1 kHz, and 20 kHz at -1 dB for five seconds, quarter channels driven 3. Long term: 1/8th power pink noise for one hour, all channels driven
	300 watts	NA	4Ω	
THD+N (1 kHz)	0.004%–0.009%		8Ω	1 dB below rated power, AES-17 20 kHz LPF, two adjacent channels driven
	0.009%–0.02%		4Ω	
THD+N (20 Hz to 20 kHz)	0.05%		8Ω	
	0.20%		4Ω	
Frequency response	20 Hz to 20 kHz, +0.4 dB/−0.2 dB		8Ω	
Intermodulation distortion ratio (SMPTE 4:1)	0.05%			1dB below rated power, SMPTE 4:1 60 Hz and 7 kHz, AES17 20 kHz LPF
Signal-to-noise ratio	109 dB			A-weighted, AES17 20 kHz LPF
Channel separation crosstalk	70 - 90 dB		8Ω	Depending upon channel utilization, measured at 1 kHz
DC offset	< ±5 mV			
Output impedance	44 mΩ			
Damping factor	180		8Ω	Measured 20 Hz to 1 kHz

Note: These specifications provide typical values and do not represent absolute limits.

A.4 CAT1416 Specifications

- Input voltage (Balanced input):
  - Nominal: 0.975 Vrms (+2 dBu)
  - Clip: 9.75 Vrms (+22 dBu)
- Input impedance: 10K Ω

A.5 Physical Specifications

Dimensions (product): 48.3 cm (19 inches) × 17.7 cm (7 inches) × 56.3 cm (22 inches)

Dimensions (shipping): 61.0 cm (24 inches) x 30.5 cm (12 inches) × 72.4 cm (28.5 inches)

DMA16302 or DMA16301 product weight: 60 lbs (27.2 kg)
shipping weight: 71.6 lbs (32.5 kg)

DMA24302 product weight: 65 ls (29.5 kg)
Shipping weight: 76.6 lbs (34.7 kg)

DMA32301 product weight: 69.6 lbs (31.57 kg)
shipping weight: 80 lbs (36 kg)

DMA16300 product weight: 54 lbs (25.5 kg)
shipping weight: 65 lbs (29.5 kg)

DMA24300 product weight: 58 lbs (26.3 kg)
shipping weight: 69 lbs (31.3 kg)

DMA32300 product weight: 62 lbs (28.1 kg)
shipping weight: 73 lbs (33.1 kg)

Operating Environmental Conditions:
Temperature range: 0°C to 40°C (32°F to 104°F)
Humidity: 20 to 80 percent relative, noncondensing

Nonoperating Environmental Conditions (storage):
Temperature range: –10°C to 60°C (14°F to 140°F)
Humidity: 20 to 80 percent relative, noncondensing

Fuse Information

B.1 Fuses

Several of the cards in the Dolby® Multichannel Amplifier contain fuses. Only Dolby authorized service technicians should test and replace these fuses after the unit has been unplugged from its power source for at least one full minute and checked to verify that the harmful voltage has been dissipated.

The Dolby Multichannel Amplifier contains dangerous voltage. Dolby provides a DMA service manual with steps that must be followed before working on the internal components of the amplifier.

We recommend that you keep these fuses on hand, so they are available for the authorized technician when needed. You can purchase replacement fuses from local suppliers (but not from Dolby Laboratories). The following table lists the Dolby Multichannel Amplifier fuses.

Table B-1 Dolby Multichannel Amplifier Fuses

Dolby Multichannel Amplifier Board	Quantity	Maximum Amps	Voltage	Blow Type	Manufacturer Part Number
CAT1422 and CAT1433	2	20 A	500 VDC	Fast Blow	Littelfuse 0505020.MXP
CAT1423	3	10 A	500 VDC	Slow blow	Littelfuse 0477010.MXP
CAT1426	2	16 A	500 VDC	Slow blow	Littelfuse 0477016.MXP
CAT1427	1	2 A	250 VAC	Slow blow	Bel Fuse Inc 5HT 2-R
CAT1428	1	30 A	500 VDC	Fast blow	Littelfuse 0505030.MXP
CAT1428	1	10 A	250 VAC	Slow blow	Schurter 0001.2514

CAT1416 Cable Pinouts

C.1 CAT1416 Male-to-Female 25-Pin D-Connector Cable Pinouts

Signal Name	Pin
Left Shield	1
Left Positive	2
Left Negative	14
Center Shield	4
Center Positive	5
Center Negative	17
Right Shield	7
Right Positive	8
Right Negative	20
Left Surround Shield	22
Left Surround Positive	23
Left Surround Negative	10
Right Surround Shield	9
Right Surround Positive	24
Right Surround Negative	11
Subwoofer Shield	13
Subwoofer Positive	25
Subwoofer Negative	12
Back Surround Left Shield	15
Back Surround Left Positive	16
Back Surround Left Negative	3
Back Surround Right Shield	18
Back Surround Right Positive	19
Back Surround Right Negative	6
Spare	-------------------

Dolby Multichannel Amplifier Part Numbers

D.1 Decoding Part Numbers

The Dolby Multichannel Amplifier (DMA) part number indicates the channel count and revision number of the unit.

- For example, in DMA *XX YY Z*, the first two digits (*XX*) indicate the number of unbridged channels in the amplifier.

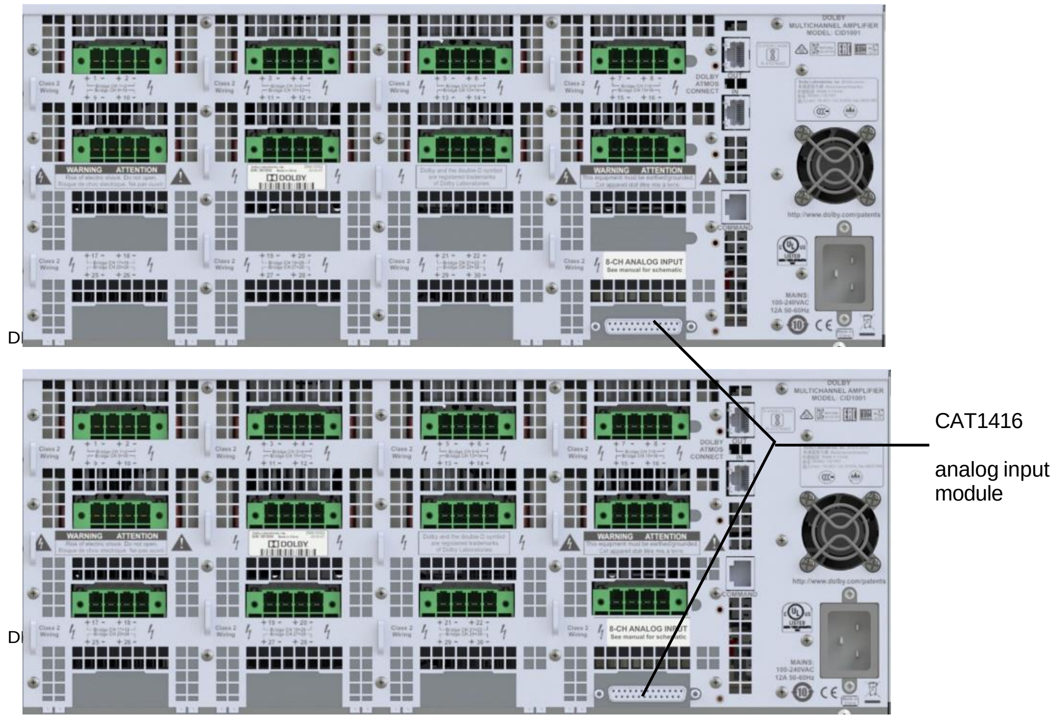
XX	Channel Count
16	16 channels
24	24 channels
32	32 channels

- The next two digits (*YY* in the example) indicate the nominal 8-ohm power output, where 30 indicates 300 W into 8-ohms.
- The last digit (*Z*) indicates the revision number.

Z	Revision Description
0	Original version of the amplifiers incorporating the CAT1422 amplifier output cards providing 300 W @ 8-ohms and 4-ohms
1	Version 2 of the amplifiers that incorporate the CAT1433 higher power amplifier output cards providing 300W @ 8-ohms and 600 W @ 4-ohms
2	Version 3 of the amplifiers that also include the CAT1433 amplifier output cards and add the CAT1416 8-channel analog input module

D.1.1 DMA16302 and DMA24302 Rear Panels

The following figures show the DMA16302 with 16 channels (unbridged) of amplification and the DMA24302 with 24 channels (unbridged) of amplification.



Note: In both of these units, the CAT1416 8-channel analog input module occupies the bottom portion of the amplifier and includes the DB25 analog input connector in the lower right section of the rear panel.

D.1.2 Power Output

The following table lists the capabilities of the two types of amplifier cards with different speaker loads.

Output Mode	Speaker Impedance	CAT1422 (DMAXX300)	CAT1433 (DMAXX301) DMAXX302)
Stereo output mode (Unbridged)	2Ω	N/A	600 W
	4Ω	300 W	600 W
	8Ω	300 W	300 W
Bridged output mode	2Ω	N/A	N/A
	4Ω	N/A	1100 W
	8Ω	600 W	1100 W

Dolby Multichannel Amplifier Gain Recommendations

E.1 Introduction

This section provides recommendations for adjusting the gain when installing a Dolby Multichannel Amplifier (DMA) in the field. The DMA shares amplification power between channels, and the unit does not have the gain controls typically provided on conventional amplifiers. There has been some confusion regarding the gain settings required to correctly install the DMA and achieve proper signal levels.

The DMA gain structure was designed so that the default gain value of 0 dB produces the maximum voltage output. For example, if you use the Dolby Audio Room Design Tool (DARDT) to design a room with DMA amplifiers, power handling is calculated with each output of the DMA set to 0 dB.

To reach the target sound pressure level (SPL) for the room during the equalization (EQ) process, DMA gains may need to be increased temporarily and you can boost channel gains up to +11 dB. However, if the system is left with some or all the DMA gains boosted, there is no protection from clipping or speaker damage when playing loud content. In addition, damage to the DMA core power supply is possible.

The Dolby Cinema Processors CP850, CP950/A, and IMS3000, (referred to as CPs in this document), provide system protection by applying limiters in Dolby Atmos Designer (DAD). However, these limiters do not consider any DMA boost above 0 dB, which is downstream from a CP. Note that cinema processors not manufactured by Dolby can be used with a DMA, but this document refers only to Dolby CPs.

E.2 Gain recommendations

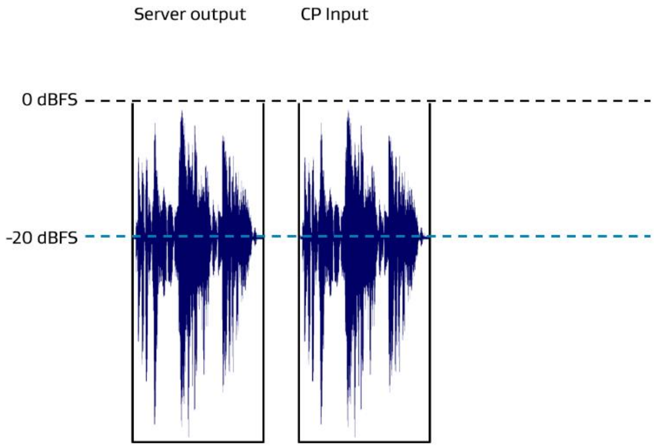
We recommend specific gain levels after completing a room EQ:

- The DMA output gain levels should be set no higher than 0 dB or as close to 0 dB as possible.
- The DMA output gain that was temporarily applied to reach the target SPL level must be applied to the CP, where the limiters can protect the system.

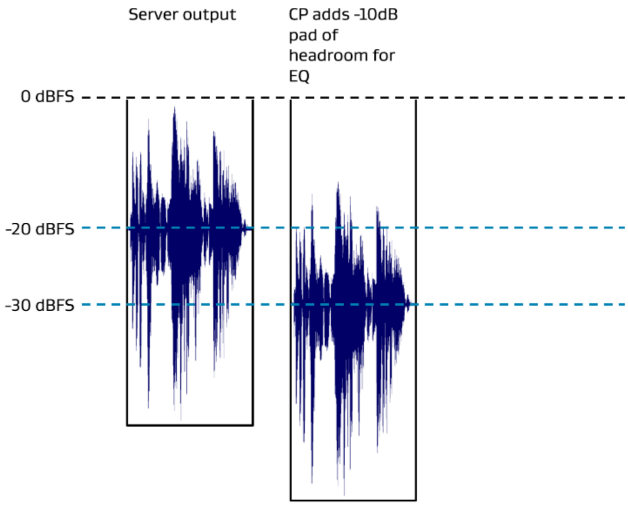
The DMA is designed to accept audio at the same reference level as a digital server/media block. In cinema, digital audio equipment is referenced at approximately -20 dBFS, which follows DCI and SMPTE standards. Dolby CPs are designed to receive audio referenced at -20 dBFS.

Following is an explanation on how a very active audio level is affected in a Dolby product workflow. The CP receives audio referenced to -20 dBFS from the server.

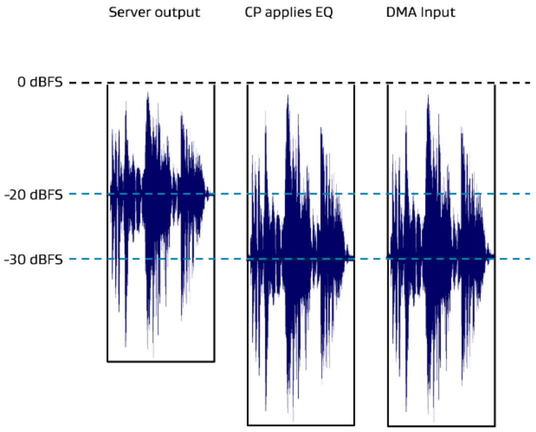
This figure approximates the dynamic range of an active audio track.



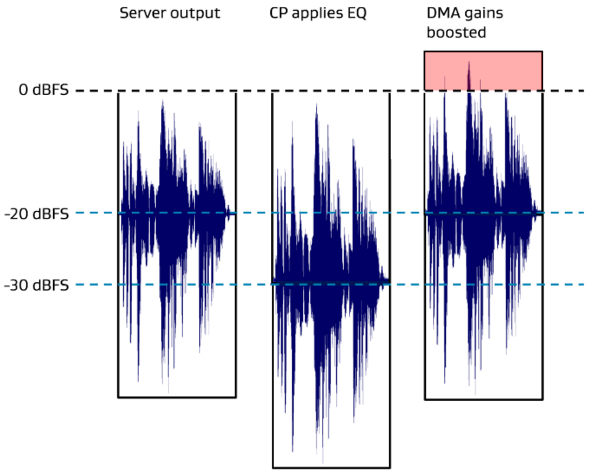
The Dolby CP adds an additional -10 dBFS pad of headroom at the input to allow more flexibility for potential EQ adjustments in the CP. As a result, the overall audio signal is referenced to a value closer to -30 dBFS.



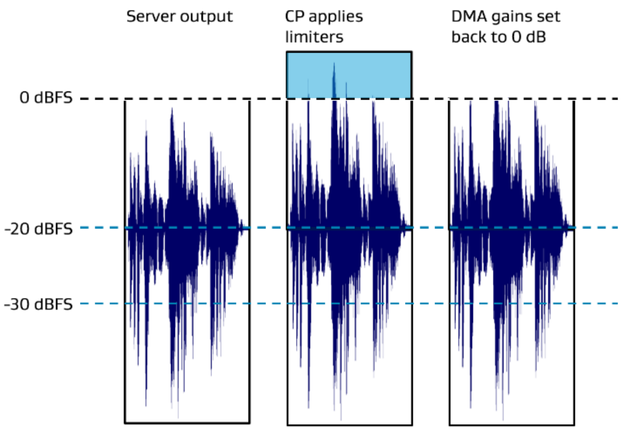
After the AutoEQ process is completed, the CP takes advantage of the increased headroom to apply the EQ, which is sent to downstream devices.



Taking this into account, it may be necessary to boost the DMA gain output to reach the reference level. This means that any peaks of very active content exceed the DMA expected output level. The red area in the following figure indicates where the hard clipping occurs in the DMA.



To protect the system, you should adjust the DMA gains back to 0 dB and increase the gains required to reach level in the CP after completing AutoEQ in DAD. This ensures that the limiters in the CP safely protect the DMA from peaks in very active content.



We recommend two methods to protect the system on the CP side:

Primary recommendation: Increase the gain in DAD after AutoEQ is completed, before pushing the configuration to the CP. This ensures that the gains are saved in the AutoEQ file and any copies of AutoEQ inherit the protections.

Secondary recommendation: Increase the CP output level gain in a custom EQ after AutoEQ is completed and the configuration is pushed. This includes editing a custom EQ preset (normally a copy of AutoEQ) in the CP web UI **equalization** screen. You can apply The EQ to all macros. There is a chance of error if that EQ preset is lost or not applied. In such a case, the levels need to be redone.

Following are basic and detailed instructions for both of these methods:

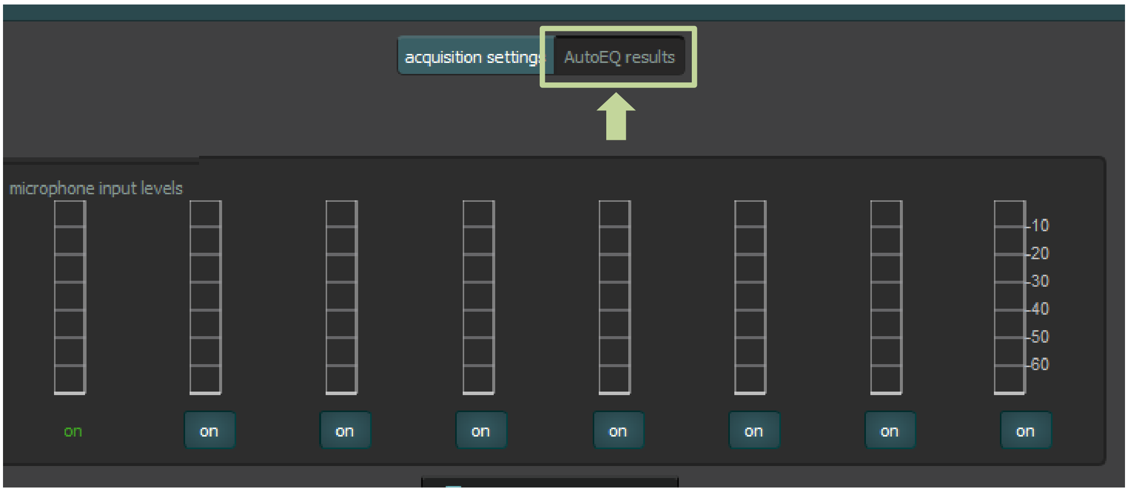
Primary basic instructions

1. Adjust the **Gain** to reach the SPL target in the DMA web UI.
2. Enter the **Amp Gain** trim values in DAD when the DMA **Gain** value is below 0 dB.
3. Perform the AutoEQ process in DAD.
4. Note any **Gain** values in the DMA web UI that are greater than 0 dB.
5. Use the **edit gain** function in DAD to apply any DMA **Gain** values that are greater than 0 dB.
6. Reset the DMA **Gain** values in the DMA web UI to 0 dB.
7. Push the AutoEQ and room configuration to the CP in DAD.

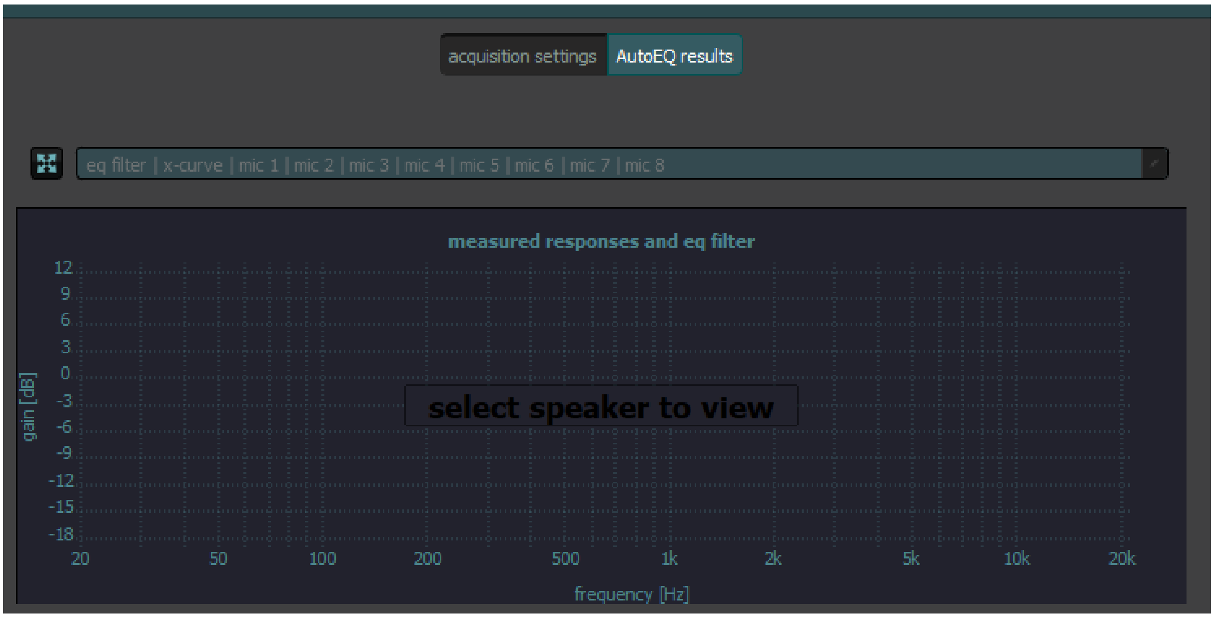
Primary detailed instructions

You can use the DAD **edit gain** function after completing AutoEQ. This ensures that the output gain is adjusted in the AutoEQ file and is applied after the configuration is pushed to the CP. The DAD **edit gain** function is nonresponsive unless the instructions are followed in order. Follow these instructions using DAD v3.2.15 or later:

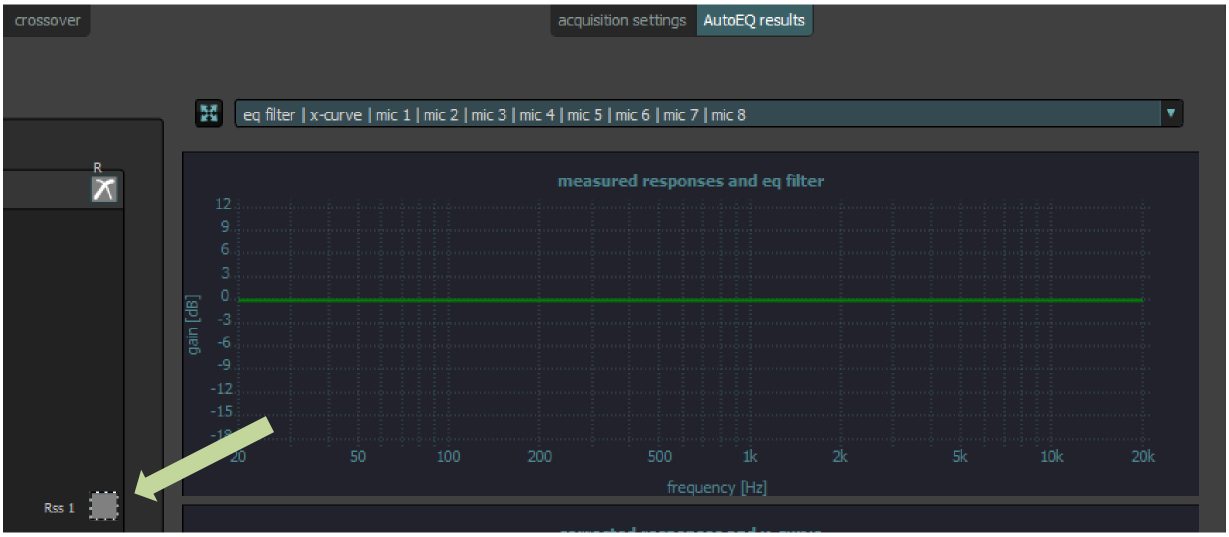
1. In the tune screen, select the **AutoEQ results** button next to **acquisition settings**.



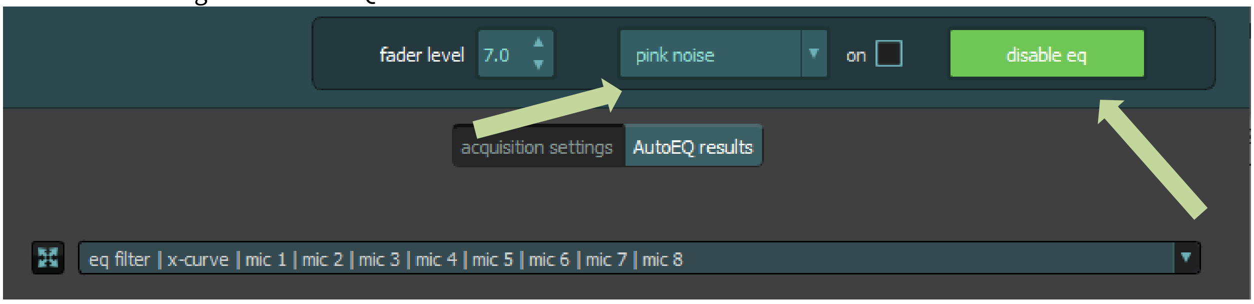
The UI changes, the **AutoEQ results** button is now highlighted, and additional controls appear.



1. Click on the first speaker you want to adjust, which is indicated by the dotted border.



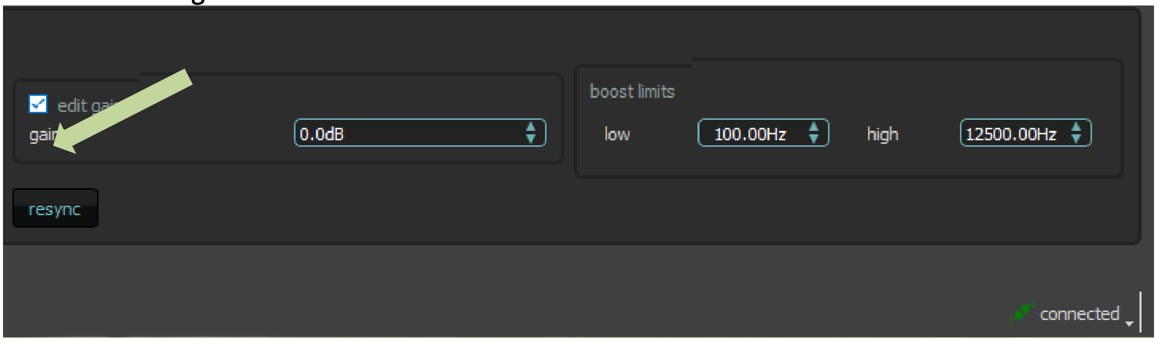
2. Click on the **enable eq** button in the upper-right corner.
   The button turns to green and its label changes to **disable eq**. The system applies the EQ, and pressing the button again disables EQ.



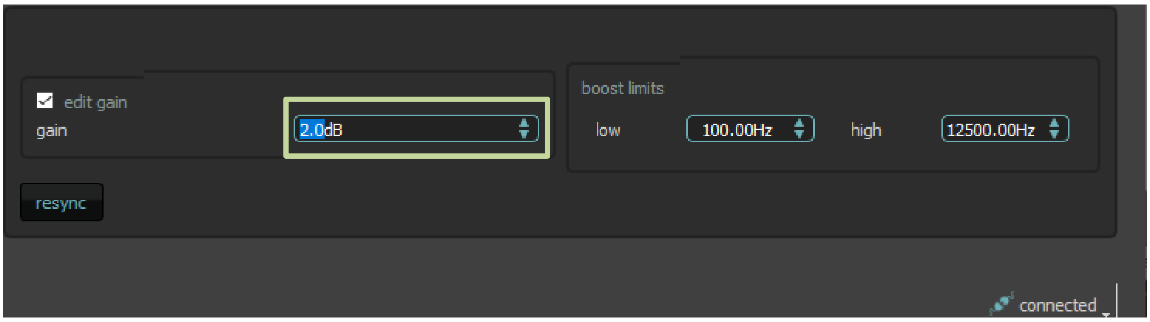
3. To the left of the **enable/disable eq** button, select **pink noise** for the test tone (pink noise is default) and check the **on** box.

It may take a minute or more for DAD to negotiate with the CP.

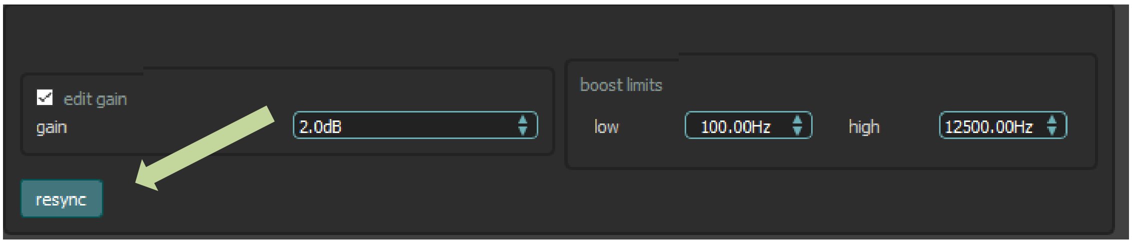
4. Click the **edit gain** check box near the bottom of the screen.



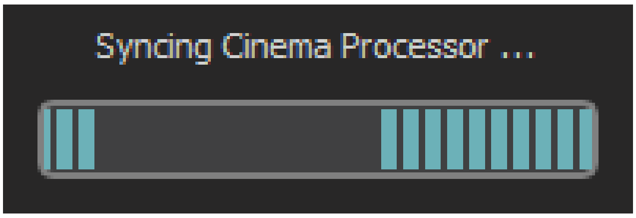
5. Click the up/down arrows or enter the desired gain value manually in the **edit gain** field.



6. Click on the **resync** button below **edit gain** to update the gain adjustment.



A small pop-up message appears for a few seconds that says **Syncing Cinema Processor…,** and then it disappears.



You can now measure the gain adjustment change in your SPL measurement. Repeat the previous procedure until you reach the desired gain level. If you are satisfied with your gain adjustment, be sure that the **edit gain** check box remains checked (do not uncheck it) before moving on to the next speaker.

Note: If you perform another speaker acquisition or update an existing acquisition, the system
unchecks the **edit gain** check box.

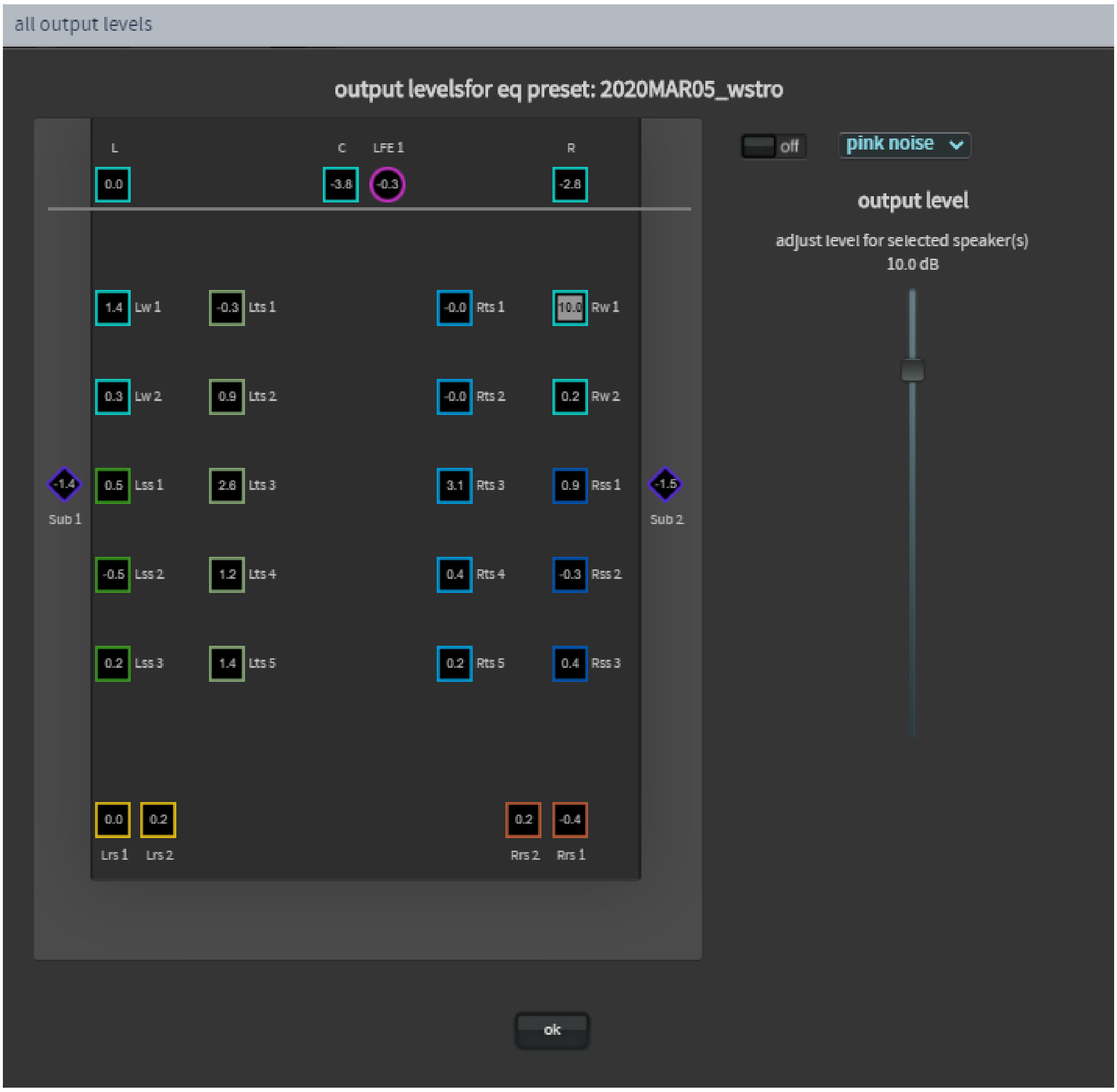
Secondary basic instructions

1. Adjust the **Gain** to reach the SPL target in the DMA web UI.
2. Enter the **Amp Gain** trim values in DAD when the DMA **Gain** value is below 0 dB.
3. Perform the AutoEQ process in DAD.
4. Push AutoEQ and the room configuration to the CP in DAD.
5. Create a new copy of AutoEQ from the **equalization** screen in the CP web UI.
6. Note any **Gain** values in the CP web UI that are greater than 0 dB, and apply to the **Gain** value for each speaker in the CP Web UI using the **edit gain** function.
7. For **Gain** values that are greater than 0 dB, reset the **Gain** values in the DMA web UI to 0 dB.
8. Apply the custom EQ in the CP web UI to all macros that use it.

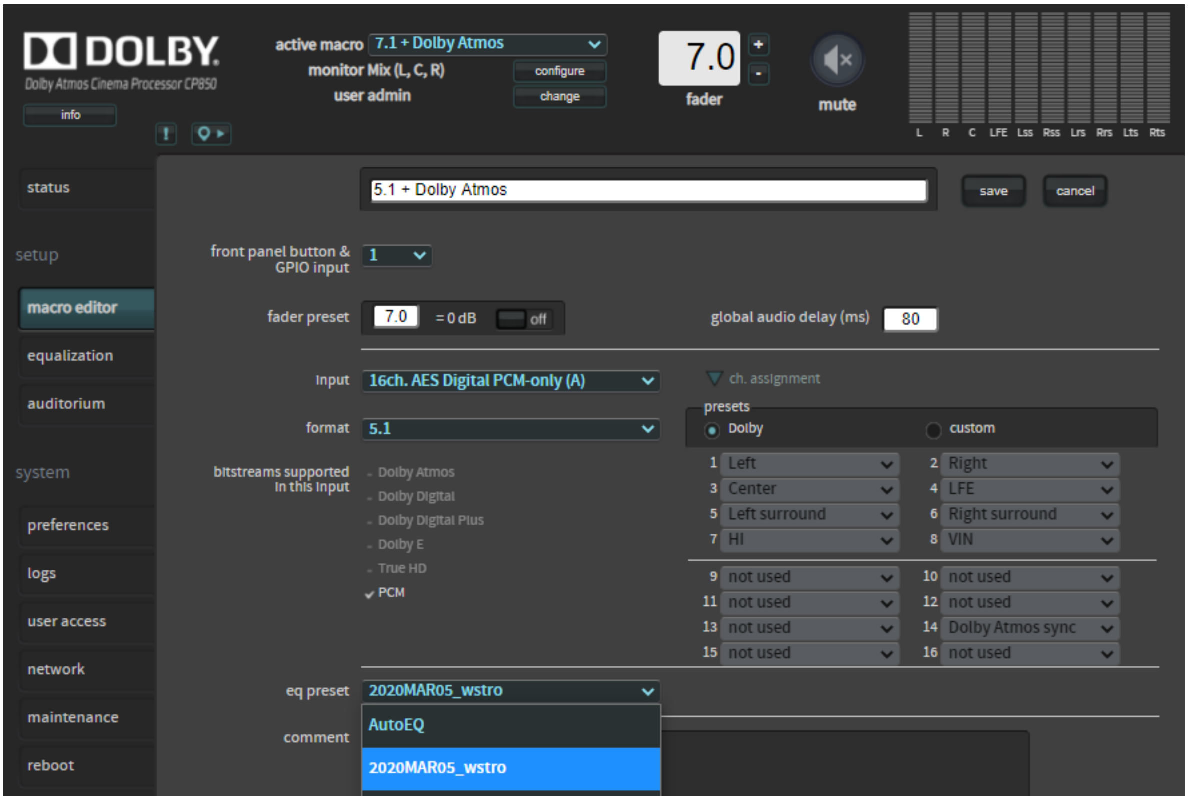
Secondary detailed instructions

You can increase the output levels in a custom EQ preset and apply it to all macros. You can do this in the **equalization** screen in the CP web UI by creating a custom EQ preset. The AutoEQ preset cannot be modified, so you need to create a new one by clicking the **+** symbol or you can edit an existing custom EQ preset.

In the **equalization** screen, you can click on the **all output levels** button to adjust each output level to the desired level.



After saving your changes, be sure to assign the custom EQ preset to all the macros in the CP web UI **macro editor**.



Note: To clarify, for **amp gain** in DAD, the DAD profile for the DMA assumes maximum output with a 0 dB gain value. Negative trim is used only in DAD when the gain value was trimmed below 0 dB. This means that a **0** value in the DAD **amp gain** menu is equal to a 0 dB gain in the DMA for that channel. We do not display boost in this menu, as in other digital amplifiers. We display boost only when the **amp gain** is nominal, which means 0, or if it was trimmed.

For example, if the DMA is overperforming at 0 dB and you reduced the DMA output channel gain by -2 dB to reach the target level, you enter -2 dB as the **amp trim** value for that output in DAD. In DAD, you may notice that amp trim is not possible for the stage speakers in the previous image.

This is because **amp trim** is not an option if you specified that the speaker should use **internal crossover**, **generic**, or **external processor** as the speaker type.